

IAL Chemistry Unit 5 (WCH15/01) — Complete Study Guide for a Near-Perfect Score

Transition Metals and Organic Nitrogen Chemistry *Pearson Edexcel International Advanced Level*

Built from analysis of 20 past papers and mark schemes, June 2019 → June 2026, with additional specification content the papers have not yet tested but could.

0. THE PAPER

Code	WCH15/01 (papers before 2020 are the legacy WCH05)
Time	1 hour 45 minutes
Total	90 marks
Section A	20 marks — multiple choice (16–20 questions, some with multiple parts)
Section B	~50 marks — structured questions
Section C	~20 marks — one long, context-based question built round a passage
Asterisked (*) questions	1–3 per paper; marked for logical structure and linked reasoning as well as content
You must have	Scientific calculator, Data Booklet , ruler

Rate of working: 90 marks in 105 minutes $\approx$ **1.17 min/mark**. The paper itself instructs you to spend **no more than 20 minutes on Section A**. That leaves ~85 minutes for 70 marks.

Topic weighting across the 20 papers

Topic	Marks per paper	Frequency

Transition metal complexes: shapes, ligands, colour, reactions	15–25	Every paper
Redox: oxidation numbers, half-equations, titration calculations	12–20	Every paper
Electrode potentials, cells, feasibility	8–17	Every paper
Arenes: benzene structure, electrophilic substitution mechanisms	10–18	Every paper
Organic nitrogen: amines, amides, amino acids, diazonium	8–16	Every paper
Spectroscopy and structure determination (NMR, IR, MS)	5–14	~90% of papers
Organic synthesis routes (Grignard, nitriles, multi-step)	5–12	~85% of papers
Polymers	3–10	~60% of papers
Catalysis (hetero-, homo-, autocatalysis)	3–8	~75% of papers
Formulae from analytical data (empirical, gas volumes, ppm)	4–10	Every paper

Two planning consequences.

1. **Transition metal chemistry plus redox/electrochemistry is 35–50 marks — over half the paper.** If your inorganic is weak you cannot reach an A*, no matter how good your organic is.
2. **Section C is 20 marks in one question.** It is always context-based and always synoptic, drawing on Units 1–4 as well as Unit 5. Never leave yourself short of time for it.

What separates near-perfect from an A

1. **Exact mark-scheme wording.** Chemistry mark schemes list *Do not award* alternatives explicitly. "The d orbitals split" scores; "the d orbital splits" is rejected (singular). "The lone pair is delocalised into the ring" scores; "the electron pair on the OH is delocalised" is rejected.
2. **Mechanism drawing precision.** Curly arrows must start from the right place and end in

the right place; the intermediate horseshoe must span **at least three carbons** and not the whole ring.

3. **The Data Booklet.** Several questions each paper are unanswerable without fluently locating electrode potentials and IR/NMR data. Practise with the booklet open from day one.
4. **Full titration calculations laid out step by step.** These are 4–6 marks each and appear in every paper, always with "give your answer to an appropriate number of significant figures".

PART A — REDOX AND ELECTROCHEMISTRY

1. Oxidation numbers

1.1 The rules

Uncombined element	= 0
Simple ion	= charge on the ion
Group 1 = +1 · Group 2 = +2 · Al = +3	
H = +1 (except -1 in metal hydrides)	
O = -2 (except -1 in peroxides, +2 in F ₂ O)	
F = -1 always	
Sum over a neutral compound	= 0
Sum over an ion	= charge on the ion

Worked examples that have appeared:

[Cr(OH) ₆] ³⁻	:	x + 6(-1) = -3	→	Cr = +3
[Cr(H ₂ O) ₆] ²⁺	:	x + 0 = +2	→	Cr = +2
Cr ₂ O ₇ ²⁻	:	2x + 7(-2) = -2	→	Cr = +6
K ₂ FeO ₄	:	2(+1) + x + 4(-2) = 0	→	Fe = +6
[Fe(CN) ₆] ⁴⁻	:	x + 6(-1) = -4	→	Fe = +2
VO ₂ ⁺	:	x + (-2) = +2	→	V = +4
VO ₂ ⁺ (dioxovanadium(V), VO ₂ ⁺)	:	x + 2(-2) = +1	→	V = +5
H ₃ MnO ₄	:	3(+1) + x + 4(-2) = 0	→	Mn = +5
[Rh(CO) ₂ I] ⁻	:	CO is neutral, I is -1 → x + 0 + (-2) = -1	→	Rh = +1

⚠ **Trap:** in complexes, work out the **charge on the metal**, not the charge on the whole ion. Neutral ligands (H₂O, NH₃, CO, en) contribute 0; anionic ligands (Cl⁻, CN⁻, OH⁻, SCN⁻,

$\text{C}_2\text{O}_4^{2-}$, EDTA^{4-}) contribute their own charge.

1.2 Using oxidation numbers for stoichiometry

A recurring MCQ and structured-question type: *"By considering oxidation numbers, how many moles of A are needed to reduce 1 mol of B?"*

Method:

1. Find the oxidation number change per atom for each species
2. Multiply by the number of atoms
3. Balance: total electrons lost = total electrons gained

Worked (June 2019 Q2): $\text{IO}_3^- \rightarrow \text{I}_2$: iodine goes $+5 \rightarrow 0$, gain of 5 e^- per I. $\text{HSO}_3^- \rightarrow \text{SO}_4^{2-}$: sulfur goes $+4 \rightarrow +6$, loss of 2 e^- per S. So 1 mol IO_3^- (5 e^- gained) needs **2.5 mol** HSO_3^- .

Worked (June 2020 Q19): in acid $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ is a 5-electron change; in neutral $\text{MnO}_4^- \rightarrow \text{MnO}_2$ is a 3-electron change. The same iron(II) requires $5/3$ as much MnO_4^- : $22.0 \times 5/3 = 36.7\text{ cm}^3$.

1.3 Proving a reaction is redox

The 2-mark answer, every time:

"[Element] changes from [oxidation number] to [oxidation number], so it is oxidised/reduced; [other element] changes from ___ to ___, so it is reduced/oxidised. Both oxidation and reduction occur, so it is a redox reaction."

Quote **numbers**, not just "it gains electrons". Mark schemes require the numbers.

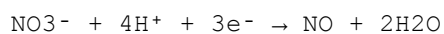
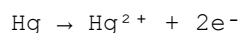
Proving a reaction is NOT redox (Jan 2025 Q20a(iv)): $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$ — chromium is **+6 on both sides**, oxygen -2 and hydrogen $+1$ throughout, so no oxidation numbers change and it is **not** a redox reaction (it is an acid–base equilibrium).

1.4 Building half-equations and full equations

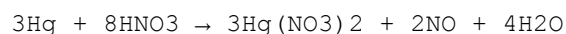
The five-step routine:

1. Write the species that change, with the correct formulae
2. Balance the atom that changes oxidation number
3. Balance O by adding H_2O
4. Balance H by adding H^+ (or, in alkali, add OH^- to both sides)
5. Balance charge by adding e^-

Worked (Oct 2023 — mercury and nitric acid):



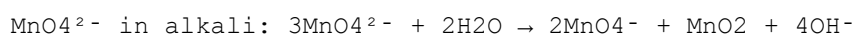
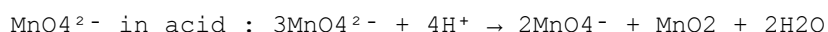
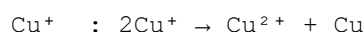
Multiply: 3 × first, 2 × second



Combining half-equations from the Data Booklet — the single most frequent structured task:

- Reverse the **oxidation** half-equation
- Multiply so the electrons cancel
- Add and cancel any species appearing on both sides

Disproportionation. The same element is **simultaneously oxidised and reduced**.



To build one: take the **half-equation in which the species is reduced** and the **reverse of the half-equation in which it is oxidised**, and combine.

2. Standard electrode potentials

2.1 The standard hydrogen electrode (SHE)

Standard conditions, all four needed:

Temperature	298 K (25 °C)
H ₂ pressure	100 kPa (1 atm)
[H ⁺]	1.00 mol dm ⁻³ (from a strong acid, e.g. 1.00 mol dm ⁻³ HCl or 0
Electrode	platinum coated with platinum black

Why platinum black? It has a **large surface area** and **catalyses the establishment of equilibrium** between H₂ and H⁺, so the potential is reached quickly and is stable. (June 2021 Q2 — the answer is *“to increase the rate of the equilibrium”*, not *“to protect the electrode”*.)

Why platinum? It is **inert** (does not react) and **conducts electricity**.

Labelling the diagram (a repeated 3-mark question): H₂ gas at 100 kPa · platinum electrode coated in platinum black · 1.00 mol dm⁻³ H⁺(aq) · temperature 298 K.

2.2 Setting up any half-cell

Half-cell type	Electrode	Solution
Metal / metal ion (e.g. Zn^{2+}/Zn)	the metal itself	1.00 mol dm^{-3} of the metal ion
Two aqueous ions (e.g. $\text{Fe}^{3+}/\text{Fe}^{2+}$)	platinum	1.00 mol dm^{-3} of both ions
Ion + gas (e.g. Cl_2/Cl^-)	platinum	1.00 mol dm^{-3} Cl^- , Cl_2 at 100 kPa
Ion + H^+ (e.g. $\text{MnO}_4^-/\text{Mn}^{2+}$)	platinum	1.00 mol dm^{-3} MnO_4^- , Mn^{2+} and H^+

⚠ **The most common lost mark:** forgetting that acidified half-cells need **1.00 mol dm^{-3} H^+** as well as the two ion species. Jan 2025 Q18a and June 2019 Q3 both test exactly this.

Making a standard $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell by mixing (June 2021 Q2c): mixing equal volumes **halves** each concentration, so you need **2 mol dm^{-3} FeSO_4** and **1 mol dm^{-3} $\text{Fe}_2(\text{SO}_4)_3$** (the latter gives 2 mol dm^{-3} Fe^{3+} , which becomes 1 after mixing).

2.3 The salt bridge

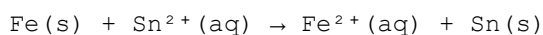
- Usually **saturated $\text{KNO}_3(\text{aq})$** (or KCl , provided no Ag^+ or Pb^{2+} is present, which would precipitate the chloride).
- Function:** completes the electrical circuit by allowing **ions to flow** between the half-cells, maintaining electrical neutrality.
- Why not a metal wire?** A wire would allow **electrons** to flow directly, and the metal would **react** with the solutions / set up its own electrode potentials.
- The ions in the bridge must **not react** with either solution.

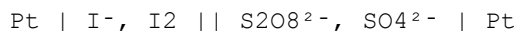
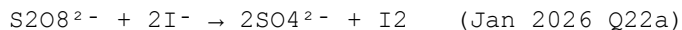
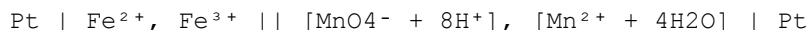
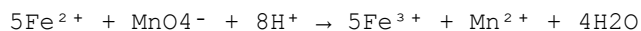
2.4 Cell diagrams

Conventions to follow exactly:

- The MOST NEGATIVE electrode system goes on the LEFT
- Single vertical line $|$ = phase boundary
- Double vertical line $||$ = salt bridge
- Species in the same phase are separated by a comma
- Working outward from the salt bridge, the OXIDISED species is written nearest

Worked examples from the papers:





Note that **platinum electrodes are needed on both sides** whenever neither half-cell contains a solid metal.

2.5 Calculating E°_{cell}

$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ(\text{right-hand electrode}) - E^\circ(\text{left-hand electrode}) \\ &= E^\circ(\text{reduction, the species being reduced}) - E^\circ(\text{the species being oxidised}) \end{aligned}$$

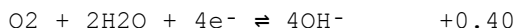
Or, more safely for beginners:

$$E^\circ_{\text{cell}} = E^\circ(\text{more positive half-cell}) - E^\circ(\text{more negative half-cell}) \quad [\text{always positive}]$$

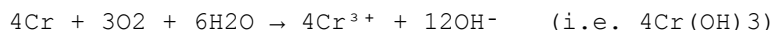
The more positive half-cell runs as REDUCTION; the more negative runs as OXIDATION

Worked examples:

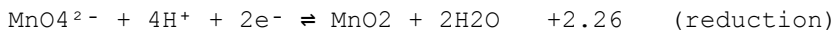
Chromium + oxygen (June 2019 Q17a):



$$E^\circ_{\text{cell}} = 0.40 - (-0.74) = +1.14 \text{ V}$$



Disproportionation of MnO_4^{2-} in acid (June 2021 Q18a):



$$E^\circ_{\text{cell}} = 2.26 - 0.56 = +1.70 \text{ V, so feasible}$$



2.6 Feasibility — what E°_{cell} does and does not tell you

The rule:

A reaction is **thermodynamically feasible** if E°_{cell} is positive.

The three linked relationships (a recurring MCQ and a 3-mark structured question):

$$\Delta S_{\text{total}} = nFE^\circ_{\text{cell}} / T \quad \rightarrow \quad E^\circ_{\text{cell}} \propto \Delta S_{\text{total}}$$

$$\begin{array}{lll} \Delta G & = & -nFE^{\circ}_{\text{cell}} \quad \rightarrow \quad E^{\circ}_{\text{cell}} \text{ positive} \Rightarrow \Delta G \text{ negative} \\ \ln K & = & \Delta S_{\text{total}} / R \quad \rightarrow \quad E^{\circ}_{\text{cell}} \propto \ln K \end{array}$$

So **E°_{cell} is directly proportional to ΔS_{total} and to $\ln K$** — *not* to ΔS_{system} and *not* to K itself. This exact distractor set appears in June 2019 Q4 and Oct 2019 Q15.

Model answer (Jan 2026 Q22b(iii), 3 marks):

E°_{cell} is directly proportional to the total entropy change. A positive E°_{cell} means ΔS_{total} is positive, and a reaction is thermodynamically feasible when the total entropy change is positive. Since $E^{\circ}_{\text{cell}} = +1.47 \text{ V}$ is positive, the reaction is feasible.

⚠ THE CRITICAL LIMITATION — a guaranteed exam point. A thermodynamically feasible reaction may still not occur, because:

1. **The activation energy may be very high**, so the rate is negligibly slow (a **kinetic barrier**).
2. The **conditions are not standard** — concentrations, temperature or pressure differ from standard values, so the actual E values differ.

Worked (Jan 2026 Q22b(iv)): $\text{S}_2\text{O}_8^{2-} + 2\text{I}^-$ is feasible but slow **because both ions are negatively charged, so they repel each other**, giving a **high activation energy**.

2.7 The effect of concentration (Le Chatelier)

Every paper asks some version of “*what happens to E if this solution is diluted?*”

The method:

1. Write the half-equation as an equilibrium
2. Decide which way the equilibrium shifts on dilution
3. Shift RIGHT (toward the products/reduction) $\rightarrow E$ becomes MORE POSITIVE
Shift LEFT $\rightarrow E$ becomes MORE NEGATIVE
4. Then work out the effect on E°_{cell}

Worked (Jan 2025 Q18c): the cell is $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Fe}^{3+}, \text{Fe}^{2+} \mid \text{Pt}$. $\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$. Diluting Zn^{2+} shifts this equilibrium to the **left**, making the zinc electrode potential **more negative**. $E^{\circ}_{\text{cell}} = E(\text{Fe}) - E(\text{Zn})$, so E°_{cell} **increases** (becomes larger than +1.53 V).

The Nernst equation is sometimes supplied in the question (June 2020 Q21c, June 2022 Q16d):

$$E = E^{\circ} + (RT / nF) \ln[M^{n+}] \quad \text{with } F = 96\,500 \text{ C mol}^{-1}, R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

Worked: for Cr^{3+}/Cr at 298 K with $[\text{Cr}^{3+}] = 0.0100 \text{ mol dm}^{-3}$: $E = -0.74 + (8.31 \times 298) / (96500 \times 3) \times \ln(0.0100) = -0.74 - 0.039 = -0.78 \text{ V}$

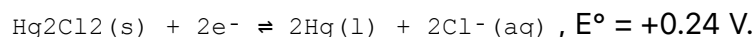
2.8 Running a cell / electrolysis-style changes

“The voltmeter is removed and the cell runs for an hour — explain the changes in the right-hand half-cell” (June 2024 Q14b(iii)):

The right-hand (more positive) electrode is where **reduction** occurs, so the concentration of the **oxidised species decreases** and the reduced species increases. If a metal is deposited, the electrode **gains mass**; if the metal dissolves at the negative electrode, that electrode **loses mass**. The colour of the solution changes accordingly, and E°_{cell} **falls towards zero** as concentrations move away from standard.

“Why might the initial voltmeter reading differ from the Data Booklet value?” (June 2024 Q14b(ii)): the conditions are **not standard** (concentration, temperature, or pressure); the solutions may be impure or the equilibrium may not yet be established.

2.9 The calomel electrode (a Section B favourite)



- **KCl crystals** are needed to keep the solution **saturated**, so that $[\text{Cl}^-]$ stays constant and the potential remains fixed.
- **Advantages over the SHE:** it is **safer** (no flammable hydrogen), **more portable and compact, easier to set up and maintain**, and does not require a gas supply at controlled pressure.
- To use it: $E^\circ_{\text{cell}} = E^\circ(\text{right}) - E^\circ(\text{left})$. If the reading is +0.37 V with calomel on the right, $0.37 = 0.24 - E^\circ(\text{Sn}^{2+}/\text{Sn})$, **so** $E^\circ(\text{Sn}^{2+}/\text{Sn}) = -0.13 \text{ V}$.

3. Fuel cells and storage cells

3.1 The hydrogen–oxygen fuel cell

	Acidic electrolyte	Alkaline electrolyte
Anode (negative, oxidation)	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$
Cathode (positive, reduction)	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
Overall	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

E°cell	+1.23 V	+1.23 V
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Learn the labels: in **any** cell, **oxidation happens at the anode**; in a **fuel cell or galvanic cell** the anode is the **negative** electrode. The fuel is always **oxidised at the anode**.

Other fuels: methanol (DMFC), ethanol, methane, hydrazine.

DMFC anode: $\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 6\text{H}^+ + 6\text{e}^-$

DMFC overall: $2\text{CH}_3\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 4\text{H}_2\text{O}$

Ethanol anode: $\text{C}_2\text{H}_5\text{OH} + 3\text{H}_2\text{O} \rightarrow 2\text{CO}_2 + 12\text{H}^+ + 12\text{e}^-$

The 6-mark * question (June 2024 Q18) — indicative content:

Fuel is supplied continuously to the anode and oxygen (air) to the cathode. At the **anode** hydrogen is **oxidised**: $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$. The **electrons flow through the external circuit**, doing work in the motor. At the **cathode** oxygen is **reduced**: $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$. The **electrolyte allows ions to move** between the electrodes, completing the circuit. **Advantages:** the only product at the point of use is **water**, so no CO_2 or NO_x pollutants; **high efficiency** because chemical energy is converted directly to electrical energy without a heat/combustion step; runs continuously while fuel is supplied, unlike a battery which needs recharging. **Disadvantages:** hydrogen is a gas that is **difficult and hazardous to store and transport** (high pressure or low temperature); the hydrogen is usually manufactured from **fossil fuels**, so CO_2 is produced elsewhere; expensive **platinum catalysts**; limited lifetime of the membrane; there is little existing refuelling infrastructure. **Another fuel:** methanol (a liquid, so easier to store, but produces CO_2 at the point of use).

4. Redox titrations and quantitative work

Every paper has at least one full titration calculation worth 4–6 marks.

4.1 The three titration systems

System	Indicator	Colour change at end point
Manganate(VII) with Fe^{2+} , $\text{C}_2\text{O}_4^{2-}$, H_2O_2	Self-indicating	colourless → first permanent pale pink

Dichromate(VI) with Fe^{2+}	Needs an indicator: diphenylamine sulfonate, barium diphenylaminesulfonate, or N-phenylanthranilic acid	e.g. green → purple/violet
Iodine–thiosulfate	Starch , added when the solution is pale straw/yellow (near the end point)	blue-black → colourless

Why dichromate needs an indicator: the colour change from **orange $\text{Cr}_2\text{O}_7^{2-}$ to green Cr^{3+}** is not sharp enough to detect the end point precisely — the green obscures it.

Why starch is added late: if added at the start, the starch–iodine complex is so stable that iodine is **adsorbed** and released only slowly, giving a **late/inaccurate end point**.

Advantages and disadvantages of KMnO_4 (Jan 2026 Q21b):

- *Advantage:* it is **self-indicating**, so no indicator is needed, and the end point is sharp.
- *Disadvantages:* it is **not a primary standard** — it cannot be weighed out accurately because it is difficult to obtain pure, is hygroscopic, and **decomposes slowly in solution** (oxidising water), so it must be **standardised** before use. The dark colour also makes the meniscus hard to read in the burette.

Which acid?

Use dilute SULFURIC acid.

NOT hydrochloric — Cl^- would be oxidised to Cl_2 by MnO_4^- , giving a HIGH titre

NOT nitric — HNO_3 is itself an oxidising agent

NOT ethanoic — it is a weak acid, so $[\text{H}^+]$ is too low

Insufficient acid (Jan 2024 Q16b(iii), June 2021 Q18b(iii)): MnO_4^- is reduced only to **MnO_2 , a brown suspension**, instead of Mn^{2+} . That is a **3-electron** rather than a 5-electron change, so **more MnO_4^- is needed** and the **titre is higher**.

4.2 The universal titration calculation layout

Write these six lines every time; each is worth a mark.

1. moles of the titrant $n = c \times V/1000$
2. mole ratio from the equation
3. moles in the aliquot (e.g. 25.0 cm^3)
4. moles in the full flask $\times (250.0 / 25.0)$
5. mass or concentration $m = n \times M$
6. percentage, and the answer to an appropriate number of significant figures

Worked in full (Jan 2025 Q21b — iron(II) sulfate in moss killer):

$n(\text{MnO}_4^-) = 0.00740 \times 17.70/1000 = 1.310 \times 10^{-4} \text{ mol}$
 Ratio $5\text{Fe}^{2+} : 1\text{MnO}_4^-$
 $n(\text{Fe}^{2+}) \text{ in } 25.0 \text{ cm}^3 = 5 \times 1.310 \times 10^{-4} = 6.549 \times 10^{-4} \text{ mol}$
 $n(\text{Fe}^{2+}) \text{ in } 250 \text{ cm}^3 = 6.549 \times 10^{-3} \text{ mol}$
 $\text{mass FeSO}_4 = 6.549 \times 10^{-3} \times 151.9 = 0.9948 \text{ g}$
 $\% \text{ by mass} = 0.9948 / 6.42 \times 100 = 15.5 \% \quad (3 \text{ s.f., matching the data})$

The iodine–thiosulfate double-titration pattern (June 2019, Oct 2019, June 2021, June 2022 all use it):

$\text{Cu}^{2+} \text{ analysis: } 2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI} + \text{I}_2 ; \quad \text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$
 so $n(\text{Cu}^{2+}) = n(\text{S}_2\text{O}_3^{2-})$ [the ratio works out 1:1]

Dichromate analysis: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow 2\text{Cr}^{3+} + 3\text{I}_2 + 7\text{H}_2\text{O}$
 so $n(\text{Cr}_2\text{O}_7^{2-}) : n(\text{S}_2\text{O}_3^{2-}) = 1 : 6$

The back-titration pattern (June 2023 Q22 — calcium ethanedioate in spinach):

1. total moles of oxidant added
2. moles of oxidant remaining, from the back titration
3. moles reacted = total - remaining
4. moles of analyte from the ratio
5. mass and percentage

4.3 Water of crystallisation and formula determination

The standard method (Jan 2025 Q20b(ii), June 2021 Q19d, Jan 2023 Q18c):

1. Take 100 g (or the given mass)
2. Find the mass of the anhydrous part and the mass of water
3. Convert each to moles
4. Divide by the smaller → the ratio gives n

Worked (Jan 2025 Q20b(ii)): heating leaves 56.74 % of the mass, so water is 43.26 %.

Anhydrous $\text{KCr}(\text{SO}_4)_2$ has $M = 283.2$. In 100 g: $56.74/283.2 = 0.2004 \text{ mol}$ anhydrous,
 $43.26/18 = 2.403 \text{ mol}$ water. Ratio = **1 : 12**, so the salt is $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$.

Empirical formula from percentage composition: divide each % by its A_r , divide through by the smallest, multiply up to whole numbers. Then $\text{molecular formula} = \text{empirical formula} \times (M_r / \text{empirical mass})$.

4.4 ppm, ppb and concentration conversions

1 ppm by mass = 1 g in 10^6 g = 1 mg dm^{-3} (for a solution of density 1.0 g cm^{-3})
1 ppb by mass = 1 $\mu\text{g dm}^{-3}$ under the same assumption

Worked (Jan 2025 Q20c(i)): $[\text{Cr}^{3+}] = 1.345 \times 10^{-7} \text{ mol dm}^{-3}$; mass per $\text{dm}^3 = 1.345 \times 10^{-7} \times 52.0 = 6.99 \times 10^{-6} \text{ g}$; 1 dm^3 of blood has mass 1000 g, so ppb = $6.99 \times 10^{-6} / 1000 \times 10^9 = 6.99 \text{ ppb}$, which **exceeds** the 4 ppb limit.

4.5 Gas-volume and gas-law calculations

$pV = nRT$ $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; p in Pa; V in m^3 ; T in K
Molar volume at r.t.p. = $24.0 \text{ dm}^3 \text{ mol}^{-1}$

Worked (June 2021 Q19a(ii)): $n = pV/RT = (103\,000 \times 157 \times 10^{-6}) / (8.31 \times 288) = 6.757 \times 10^{-3} \text{ mol}$; $M = 0.493 / 6.757 \times 10^{-3} = 73.0 \text{ g mol}^{-1}$.

Gaseous hydrocarbon combustion by volume (June 2020 Q23a, June 2021 Q22, June 2024 Q19d). Use Avogadro: **volumes are in the same ratio as moles**.



Volume contraction on cooling = volume of reactants – volume of gases after
(water condenses, so it is not counted)
 CO_2 is absorbed by KOH / NaOH solution, so the further decrease gives $V(\text{CO}_2)$

Worked (June 2021 Q22): $25 \text{ cm}^3 \text{ C}_x\text{H}_y$; total contraction 75 cm^3 ; $\text{CO}_2 = 100 \text{ cm}^3$, so $x = 100/25 = 4$. Contraction = $(25 + V_{\text{O}_2}) - (100) = 75 \rightarrow V_{\text{O}_2} = 150 \rightarrow x + y/4 = 150/25 = 6 \rightarrow y/4 = 2 \rightarrow y = 8$. So C_4H_8 .

PART B — TRANSITION METALS

5. Definitions and electronic structure

5.1 The two definitions (learn word for word)

- **d-block element:** an element whose **highest-energy electron enters a d subshell** (in the ground-state atom, following the Aufbau principle).
- **Transition element:** a d-block element that forms **at least one stable ion with a**

partially (incompletely) filled d subshell.

Therefore Sc and Zn are d-block but NOT transition elements:

- Sc forms only Sc^{3+} , which is **$3d^0$** — the d subshell is **empty**.
- Zn forms only Zn^{2+} , which is **$3d^{10}$** — the d subshell is **full**.
- Similarly **mercury** (Jan 2024 Q13 / Aug 2023 Q21a): Hg is d-block because the last electron enters a 5d orbital, but Hg^+ and Hg^{2+} both have **completely full 5d orbitals**, so it is not a transition element.

⚠ Mark-scheme warning: the answer must be given in terms of the **ion**, not the atom.

“Zinc has a full 3d subshell” does **not** score; “the Zn^{2+} ion has a completely full 3d subshell” does.

5.2 Electronic configurations

Sc	$[\text{Ar}] 3d^1 4s^2$	Ti	$[\text{Ar}] 3d^2 4s^2$	V	$[\text{Ar}] 3d^3 4s^2$
Cr	$[\text{Ar}] 3d^5 4s^1 \leftarrow$	Mn	$[\text{Ar}] 3d^5 4s^2$	Fe	$[\text{Ar}] 3d^6 4s^2$
Co	$[\text{Ar}] 3d^7 4s^2$	Ni	$[\text{Ar}] 3d^8 4s^2$	Cu	$[\text{Ar}] 3d^{10} 4s^1 \leftarrow$
Zn	$[\text{Ar}] 3d^{10} 4s^2$				

Cr and Cu are the exceptions — a half-full ($3d^5$) or full ($3d^{10}$) d subshell is more stable, so one 4s electron is promoted.

Forming ions: the 4s electrons are removed FIRST.

Fe	$[\text{Ar}] 3d^6 4s^2$	$\rightarrow$	Fe^{2+}	$[\text{Ar}] 3d^6$	$\rightarrow$	Fe^{3+}	$[\text{Ar}] 3d^5$
Cu	$[\text{Ar}] 3d^{10} 4s^1$	$\rightarrow$	Cu^+	$[\text{Ar}] 3d^{10}$	$\rightarrow$	Cu^{2+}	$[\text{Ar}] 3d^9$
Cr	$[\text{Ar}] 3d^5 4s^1$	$\rightarrow$	Cr^{3+}	$[\text{Ar}] 3d^3$			
V	$[\text{Ar}] 3d^3 4s^2$	$\rightarrow$	V^{2+}	$[\text{Ar}] 3d^3$			

Box diagrams: fill singly with parallel spins first (Hund’s rule), then pair.

Unpaired electrons (June 2021 Q5): Cr atom has **six** ($3d^5 4s^1$) — the most of any element in the row.

5.3 Ionisation energies

Why copper’s 1st IE is lower than zinc’s (Oct 2019 Q21b(i)):

The electron removed from copper comes from the **4s** orbital, which is **further from the nucleus** and **more shielded**; zinc has **one more proton**, so a **greater nuclear charge** attracting the outer 4s electrons.

Why copper's 2nd IE is greater than zinc's:

Copper's second electron is removed from a **3d** orbital, which is **closer to the nucleus and less shielded** than zinc's 4s orbital, so more energy is needed.

Why variable oxidation states occur (a recurring 2–3 mark question):

The **4s and 3d orbitals are very close in energy**, so successive ionisation energies increase only **gradually**; there is no large jump, so **different numbers of electrons can be removed** and the energy released on forming bonds or hydrating the ion compensates for the ionisation energy.

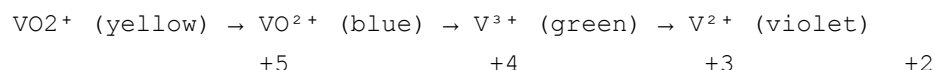
Deducing the maximum oxidation state (June 2020 Q16, Oct 2022 Q19a): the maximum equals the **total number of 4s + 3d electrons**. For $[\text{Ar}]3d^54s^1$ (chromium) that is **+6**; for vanadium $[\text{Ar}]3d^34s^2$, **+5**.

Explaining relative stability of Fe(III) over Fe(II) (June 2021 Q21a): Fe^{3+} is **$3d^5$** , a **half-filled** d subshell, which is a particularly stable arrangement; Fe^{2+} is **$3d^6$** .

5.4 The common oxidation states and their species

Metal	States	Key species and colours
Ti	+2, +3, +4	$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ purple/violet ; TiO^{2+} colourless; TiO_2 white
V	+2, +3, +4, +5	V^{2+} violet ; V^{3+} green ; VO^{2+} blue ; VO_2^+ yellow
Cr	+2, +3, +6	Cr^{2+} blue ; $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ green (violet when pure); CrO_4^{2-} yellow ; $\text{Cr}_2\text{O}_7^{2-}$ orange
Mn	+2, +4, +6, +7	Mn^{2+} very pale pink ; MnO_2 brown/black ; MnO_4^{2-} green ; MnO_4^- purple
Fe	+2, +3	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ pale green ; $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ yellow/brown
Co	+2, +3	$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ pink ; $[\text{CoCl}_4]^{2-}$ blue ; $[\text{Co}(\text{NH}_3)_6]^{2+}$ straw/yellow ; $[\text{Co}(\text{NH}_3)_6]^{3+}$ brown
Ni	+2	$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ green ; $[\text{Ni}(\text{NH}_3)_6]^{2+}$ blue/violet
Cu	+1, +2	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ blue ; $[\text{CuCl}_4]^{2-}$ yellow-green ; $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ deep royal blue ; Cu^+ compounds colourless/white
Zn	+2	all colourless / white

The vanadium reduction series (Jan 2023 Q18e — a classic * question):



Reduce with **zinc in acid**. To reduce all the way to V^{2+} you need a reducing agent whose E° is more negative than each successive step (Zn^{2+}/Zn is -0.76 V). To stop at V^{3+} , choose a reagent with an E° between the relevant values — this is exactly what the calculation questions ask.

Choosing a reducing agent: compare E° values. The reagent must give a **positive E°_{cell}** for the step you want and a **negative E°_{cell}** for the step you do not want (Oct 2022 Q19b, June 2024 Q14c).

6. Complexes: ligands, shapes and isomerism

6.1 Definitions

- **Ligand:** *a species with a lone pair of electrons that forms a **dative covalent (coordinate) bond** to a central metal ion.* (It is an **electron-pair donor** — it does **not** have to be negatively charged, which is why NH_4^+ , having no lone pair, cannot be a ligand.)
- **Complex ion:** a central metal ion surrounded by ligands bonded by dative covalent bonds.
- **Coordination number:** *the number of dative covalent bonds formed to the central metal ion* — **not** the number of ligands (a bidentate ligand forms two bonds).
- **Monodentate:** forms **one** dative bond. **Bidentate:** forms **two**. **Multidentate/hexadentate:** forms several.

⚠ **Coordination number trap:** $[\text{Ni}(\text{en})_3]^{2+}$ has three ligands but coordination number **6**; $[\text{Ni}(\text{EDTA})]^{2-}$ has one ligand but coordination number **6**; $[\text{Ni}((\text{CH}_3)_2\text{NCS}_2)_3]^-$ has three bidentate ligands, so coordination number **6**.

6.2 The ligands you must know

Monodentate	Bidentate	Hexadentate
H_2O , NH_3 , Cl^- , CN^- ,	1,2-diaminoethane (en) $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ ·	

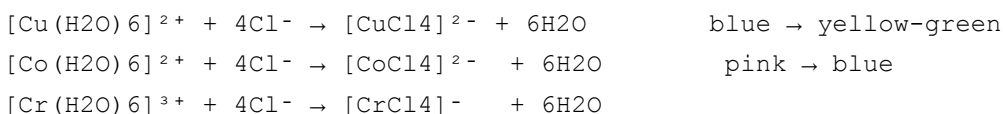
OH ⁻ , CO, SCN ⁻ , CH ₃ NH ₂ , THF	ethanedioate C ₂ O ₄ ²⁻ · benzene-1,2-diol · curcumin anion · dimethyldithiocarbamate	EDTA⁴⁻ (4 O ⁻ + 2 N)
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How to identify a bidentate ligand (Jan 2025 Q20c(ii), Oct 2024 Q18d(ii)): it has **two atoms bearing lone pairs**, far enough apart to bond to the **same metal ion** (typically 1,2- or adjacent positions), forming a stable **five-membered ring**.

6.3 Shapes

Coordination number	Shape	Bond angle	Examples
6	octahedral	90°	most aqua ions [M(H ₂ O) ₆] ⁿ⁺ , [Fe(CN) ₆] ³⁻ , [Cr(NH ₃) ₆] ³⁺ , [Ni(EDTA)] ²⁻
4	tetrahedral	109.5°	[CuCl ₄] ²⁻ , [CoCl ₄] ²⁻ , [CrCl ₄] ⁻ , [FeCl ₄] ⁻
4	square planar	90°	[Pt(NH ₃) ₂ Cl ₂] (cis-platin), [Ni(CN) ₄] ²⁻ , [Rh(CO) ₂ I ₂] ⁻ , [Au(curc) ₂] ⁺
2	linear	180°	[Ag(NH ₃) ₂] ⁺ , [Cu(NH ₃) ₂] ⁺

The chloride rule. When **concentrated HCl** (or excess Cl⁻) is added to an aqueous transition metal ion, the six water ligands are replaced by **four chloride ligands** to give a **tetrahedral** complex.



Why the coordination number falls from 6 to 4 (June 2021 Q21f, Oct 2022 Q22b(iii)):

Chloride ions are **larger** than water molecules, so only **four** can fit around the metal ion — there is not enough room for six. (Ligand repulsion; steric crowding.)

6.4 Isomerism in complexes

Cis–trans (geometric) isomerism

- In **square planar** MA₂B₂ complexes: **cis** has the two identical ligands adjacent (90°); **trans** has them opposite (180°). *Cis*-platin is anticancer active; *trans*-platin is not.

- In **octahedral** MA4B2: cis (adjacent) and trans (opposite).
- In octahedral MA3B3: **facial** (*fac*) and **meridional** (*mer*).

Optical isomerism

- Occurs when the complex has **no plane of symmetry** and is **non-superimposable on its mirror image**.
- Classic case: an octahedral complex with **three bidentate ligands**, e.g. $[\text{Ni}(\text{en})_3]^{2+}$, $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Co}(\text{en})_3]^{3+}$.
- Also **cis-MA2B2C2** octahedral complexes (the trans forms are not chiral).
- A **tetrahedral** complex with **four different ligands**, e.g. $[\text{Co}(\text{NH}_3)\text{ClBrI}]$, is chiral (Jan 2024 Q17d).
- Detected by rotating the plane of **plane-polarised light** in opposite directions.

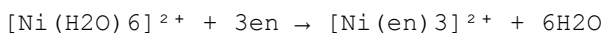
Drawing 3D structures — the marks come from the conventions:

- **Octahedral**: four ligands in the plane (two normal lines, one wedge, one dashed), plus one above and one below.
- **Tetrahedral**: two normal lines, one wedge, one dashed.
- **Square planar**: all four in the plane at 90°.
- **Always show the overall charge** outside a square bracket.

6.5 The chelate effect (a guaranteed explanation question)

The question: *why is a complex with a bidentate or multidentate ligand more stable than one with monodentate ligands?*

The model answer (3 marks):



4 particles → 7 particles, so there is an **increase in the number of moles/particles** and hence a **large positive ΔS_{system}** . The bonds broken (Ni–O) and made (Ni–N) are **similar in number and strength**, so **ΔH is approximately zero** and hence $\Delta S_{\text{surroundings}}$ is close to zero. Therefore **ΔS_{total} is positive**, so the reaction is **thermodynamically feasible** and the chelate complex is more stable.

Why EDTA⁴⁻ is preferred to en for treating metal poisoning (Jan 2025 Q20c(ii)): EDTA is **hexadentate**, so **one** EDTA molecule replaces **six** water ligands — $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + \text{EDTA}^{4-}$

→ $[\text{Cr}(\text{EDTA})]^- + 6\text{H}_2\text{O}$ gives **2 particles → 7 particles**, an even larger entropy increase and hence a **more stable** complex.

7. The colour of transition metal complexes

This is examined in **every single paper** and carries 2–6 marks. Learn the three sentences exactly.

7.1 Why complexes are coloured — the three marking points

1. The **ligands cause the (3)d orbitals to split** into two groups of slightly different energy (three lower, two upper).
2. **Visible light (a photon) is absorbed**, promoting an **electron from the lower to the higher energy level (a d–d transition)**; the energy absorbed is $\Delta E = hf$.
3. The light that is **not absorbed is transmitted or reflected**, so the **complementary colour** is seen.

⚠ Mark-scheme rejections you must avoid:

- “**d orbital** splits” (singular) — **rejected**. Always write **d orbitals** or **d subshell**, plural.
- “d–d transitions occur” on its own — only credited **if you have already said the orbitals are split**.
- “electrons fall back down and emit light” — **rejected**.
- “the colour is absorbed” — ignored; say **light/photons** are absorbed.

7.2 Why some complexes are colourless

Because the ion has **no d electrons (d^0)** or a **completely full d subshell (d^{10})**, so **electrons cannot be promoted** between the split d orbitals — **no d–d transition is possible**.

Colourless because d^0	Colourless because d^{10}
Sc^{3+} , Ti^{4+} , V^{5+} (as VO_2^+ ion this is actually yellow — see below), Cr^{6+}	Zn^{2+} , Cu^+ , Ag^+ , Ce^{3+} ($4f^1$, effectively no d–d)

⚠ **Mark-scheme rejections:** “the d orbitals cannot be split” — **rejected** (they still split, there are just no electrons to promote). “the subshell is empty” for a d^{10} ion — **rejected**.

Model answer for Zn^{2+} (Oct 2019 Q21b(iii), 2 marks):

The Zn^{2+} ion has a **completely full 3d subshell ($3d^{10}$)**, so **no electron can be promoted** from a lower to a higher d level, and there are **no d–d transitions in the visible region**.

Why is $\text{Cr}_2\text{O}_7^{2-}$ orange if Cr^{6+} is d^0 ? Because its colour arises from **charge transfer** (electron transfer from oxygen to the metal), not from d–d transitions. The same applies to MnO_4^- . This is why “if $[\text{Cr}(\text{H}_2\text{O})_6]^{6+}$ existed it would be colourless” is the expected answer in June 2019 Q17d(iii) — the aqua ion would have no charge-transfer ligand.

7.3 Why two complexes have different colours

Different **ligands** (or a different **oxidation state**, **coordination number** or **shape**) cause a **different degree of splitting** of the d orbitals. So a **different energy — and hence a different frequency/wavelength — of light is absorbed**. So a **different complementary colour is transmitted/reflected** and observed.

⚠ **Mark-scheme rejections:** “the number of ligands is different” or “the charge on the ion is different” on their own — **rejected**; you must link the difference to **the size of the splitting**.

7.4 The colour wheel

Colour absorbed	Colour seen (complementary)
red (~700 nm)	green
orange	blue
yellow (~580 nm)	violet/blue
green (~510 nm)	red/purple
blue (~470 nm)	orange
violet (~410 nm)	yellow

Questions asking “which wavelength is absorbed by this coloured solution?” are answered by taking the **complementary** colour of the observed one.

7.5 Colorimetry

- Choose a filter of the **complementary colour** to the solution (i.e. the colour the solution **absorbs**) so absorbance is maximised and sensitivity is greatest.

- Plot a **calibration curve** of absorbance against concentration using standard solutions.
- **Zero the colorimeter with a blank** (pure solvent) first.
- Read the unknown concentration off the calibration line.
- $\text{absorbance} = 2 - \log(\% \text{ transmittance})$ when transmittance data is given.
- **Limitation:** readings outside the range of the calibration data require **extrapolation**, which is unreliable because the relationship may not remain linear at high concentration.

8. Reactions of aqueous transition metal ions

These tables are the backbone of Section B. Learn them until you can reproduce them cold.

8.1 The three reaction types

Type	What happens	Example
Ligand exchange (substitution)	one ligand replaces another	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightarrow [\text{CuCl}_4]^{2-} + 6\text{H}_2\text{O}$
Deprotonation (acid–base)	a base removes H^+ from a coordinated water ligand	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{OH}^- \rightarrow [\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2] + 2\text{H}_2\text{O}$
Redox	the oxidation state of the metal changes	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+} \rightarrow [\text{Fe}(\text{H}_2\text{O})_6]^{3+} + \text{e}^-$

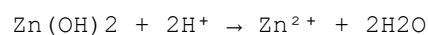
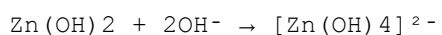
⚠ **The most-tested distinction:** adding NaOH or NH_3 to make a **precipitate** is **deprotonation**, not ligand exchange. Adding **excess** NH_3 to dissolve the precipitate **is** ligand exchange. June 2021 Q8 tests this exact pair.

8.2 With sodium hydroxide

Ion	Limited NaOH → precipitate	Excess NaOH
$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ blue	pale blue $[\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2]$	insoluble
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ pale green	green $[\text{Fe}(\text{H}_2\text{O})_4(\text{OH})_2]$, turns brown on standing (oxidised by air)	insoluble
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$		

yellow	brown $[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3]$	insoluble
$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ pink	blue $[\text{Co}(\text{H}_2\text{O})_4(\text{OH})_2]$, turns pink then brown in air	insoluble
$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ green	green $[\text{Ni}(\text{H}_2\text{O})_4(\text{OH})_2]$	insoluble
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ green	grey-green $[\text{Cr}(\text{H}_2\text{O})_3(\text{OH})_3]$	dissolves → $[\text{Cr}(\text{OH})_6]^{3-}$ green
$[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ colourless	white $[\text{Zn}(\text{H}_2\text{O})_4(\text{OH})_2]$	dissolves → $[\text{Zn}(\text{OH})_4]^{2-}$ colourless

Amphoteric — Cr(III) and Zn(II) hydroxides dissolve in **both** excess NaOH and dilute acid. “Amphoteric” is the term asked for in Oct 2019 Q21d(i).



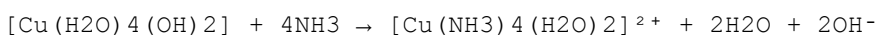
8.3 With aqueous ammonia

Limited NH₃ gives the same hydroxide precipitates as NaOH (NH₃ acts as a **base**, deprotonating the water ligands).

Excess NH₃ (NH₃ now acts as a **ligand**):

Ion	Excess NH ₃
Cu^{2+}	deep royal blue $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ — only four waters are replaced
Co^{2+}	straw/yellow $[\text{Co}(\text{NH}_3)_6]^{2+}$, oxidised in air to brown $[\text{Co}(\text{NH}_3)_6]^{3+}$
Ni^{2+}	blue/violet $[\text{Ni}(\text{NH}_3)_6]^{2+}$
Zn^{2+}	colourless $[\text{Zn}(\text{NH}_3)_4]^{2+}$
Cr^{3+}	purple $[\text{Cr}(\text{NH}_3)_6]^{3+}$
Fe^{2+} , Fe^{3+}	no further reaction — the hydroxide precipitates remain

The copper equation to memorise:



8.4 With other reagents

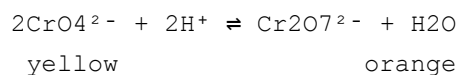
Reagent	Result
Concentrated HCl	tetrahedral chloro complexes (see §6.3)
SCN⁻ with Fe³⁺	blood-red $[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}$ — a sensitive test for Fe(III); ligand exchange
KI with Cu²⁺	$2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI}(\text{s}) + \text{I}_2(\text{aq})$ — white CuI precipitate in a brown iodine solution (redox)
Na₂CO₃ with M²⁺	metal carbonate precipitate (e.g. CuCO ₃ blue-green)
Na₂CO₃ with M³⁺	effervescence (CO₂) and a hydroxide precipitate, because $[\text{M}(\text{H}_2\text{O})_6]^{3+}$ is acidic enough to react with carbonate
Excess NH₂CH₂CH₂NH₂ (en) with Fe³⁺	$[\text{Fe}(\text{en})_3]^{3+}$
H₂O₂ / OH⁻ with Cr³⁺	oxidised to yellow CrO ₄ ²⁻
Cobalt chloride paper	anhydrous blue → hydrated pink (test for water)

8.5 The acidity of aqueous M^{3+} (and M^{2+}) ions

Why is $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ acidic? (asked repeatedly, e.g. Jan 2025 Q21c)

The metal ion has a **high charge density** (high charge, small radius), so it **polarises the O–H bonds** of the coordinated water molecules, **weakening** them. A water ligand therefore **releases an H⁺ ion** to the solution: $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+} + \text{H}_3\text{O}^+$ so the solution is acidic. M^{3+} ions are **more acidic than M^{2+}** ions because of their higher charge density.

8.6 The chromate(VI) / dichromate(VI) equilibrium



- Add **acid** → shifts right → **orange**.

- Add **alkali** → removes H^+ → shifts left → **yellow**.
- **Not a redox reaction** — chromium is +6 throughout.

9. Catalysis

9.1 Why transition metals are good catalysts

Because they have **variable oxidation states** (so they can accept and donate electrons, forming intermediates), and because they have **partially filled d orbitals** which allow reactant molecules to be **adsorbed** onto the surface.

9.2 Heterogeneous catalysis

Definition: the catalyst is in a **different phase** from the reactants.

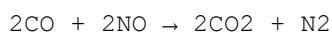
Mechanism — the four steps:

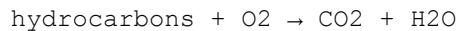
1. Reactant molecules are ADSORBED onto ACTIVE SITES on the catalyst surface
2. The bonds within the reactants are WEAKENED (and the molecules are held in the correct orientation, and in higher local concentration)
3. Reaction occurs, with a LOWER ACTIVATION ENERGY
4. Product molecules are DESORBED from the surface

Examples you must know:

Catalyst	Reaction
Fe	Haber process, $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
V2O5	Contact process, $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$
Pt / Pd / Rh	catalytic converter
Ni	hydrogenation of alkenes/vegetable oils
TiCl3 / Ziegler–Natta	polymerisation of propene

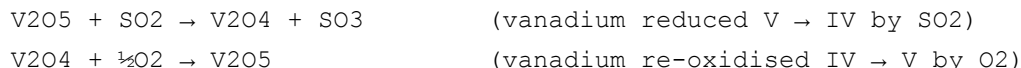
The catalytic converter: CO, unburnt hydrocarbons and NO_x are adsorbed onto the Pt/Rh surface; the reactions occur:





then the products are desorbed. **⚠ Note:** it is **not true** that the products are harmless — CO₂ is a greenhouse gas (Oct 2021 Q8 tests this).

The Contact process is a special case. Although V₂O₅ is heterogeneous, its mechanism is written as a **redox cycle**:



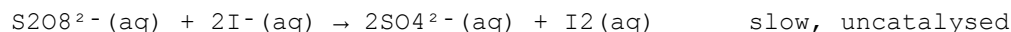
The order matters — SO₂ reduces the vanadium **first** (June 2019 Q7).

9.3 Homogeneous catalysis

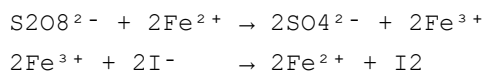
Definition: the catalyst is in the **same phase** as the reactants (usually all in aqueous solution).

Mechanism: the catalyst forms an **intermediate** by being **oxidised (or reduced)** to a **different oxidation state**, then **reverts to its original oxidation state**, providing an **alternative route with a lower activation energy**.

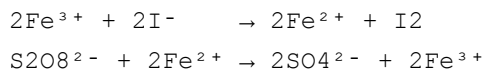
The iodide–peroxodisulfate reaction — learn both equations:



Catalysed by Fe²⁺:



Catalysed by Fe³⁺ — the same two steps, in the opposite order:

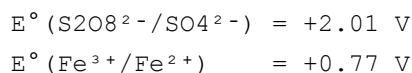


So both Fe²⁺ and Fe³⁺ catalyse it (Oct 2021 Q10).

Why the uncatalysed reaction is slow:

Both S₂O₈²⁻ and I⁻ are **negatively charged**, so they **repel** each other; a high energy is needed for them to collide successfully, giving a **high activation energy**.

Justifying the catalysis using E° values (Jan 2026 Q22c(ii), 3 marks):



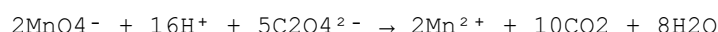
$$E^{\circ}(\text{I}_2/\text{I}^-) = +0.54 \text{ V}$$

The $\text{Fe}^{3+}/\text{Fe}^{2+}$ value lies **between** the other two, so **both** steps have a **positive E°_{cell}** and are therefore feasible: step 1 $E^{\circ}_{\text{cell}} = 2.01 - 0.77 = +1.24 \text{ V}$; step 2 $E^{\circ}_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V}$. **This is the general rule for choosing a homogeneous catalyst: its E° must lie between those of the two half-cells.**

9.4 Autocatalysis

Definition: a **product** of the reaction acts as the **catalyst**.

The example: manganate(VII) with ethanedioate ions.



The Mn^{2+} produced is the autocatalyst.

Explaining the rate–time graph (Oct 2025 Q12d(ii), 3–4 marks):

Initially the rate is slow, because there is no catalyst and the two negative ions repel each other, giving a high activation energy. As **Mn^{2+} is produced**, it **catalyses** the reaction, so the **rate increases** even though the reactant concentrations are falling. **Later the rate decreases** because the **reactants are used up**, so their concentrations — and hence the collision frequency — fall.

The concentration-of-product against time graph has a characteristic **S shape** (slow, then steep, then levelling off).

9.5 Haemoglobin and carbon monoxide

- Haemoglobin contains an **Fe(II)** ion in a **porphyrin** ring, six-coordinate; oxygen binds **reversibly** as a ligand.
 - CO poisoning:** carbon monoxide forms a **stronger dative covalent bond** with the Fe(II) than oxygen does, so the binding is effectively **irreversible**; the haemoglobin can no longer transport oxygen. (Oct 2021 Q5 — the answer is the *strength of the dative bond*, not “it binds irreversibly” alone.)
 - Why oxyhaemoglobin (red) and oxyhaemocyanin (blue) differ in colour:** they contain **different metal ions** (Fe vs Cu) with different d-electron configurations and different ligands, so the **d-orbital splitting differs**, so **different frequencies of light are absorbed** and different complementary colours are seen.
-

PART C — ARENES

10. The structure of benzene

10.1 The delocalised model

Benzene is a **planar, regular hexagon** with bond angles of **120°**. Each carbon is **sp² hybridised**, forming three **σ bonds** (two to carbons, one to hydrogen). Each carbon has **one electron in a p orbital perpendicular to the plane** of the ring. These six p orbitals **overlap sideways** to form a **delocalised π system** — rings of electron density **above and below** the plane. There are **6 delocalised electrons**, and all six C–C bonds are **identical**.

⚠ **Mark-scheme point:** the delocalised electrons come from **p orbitals** overlapping to form **π bonds** — not s orbitals, not σ bonds (June 2020 Q2, June 2021 Q10).

10.2 The four pieces of evidence (a guaranteed 4–6 mark question)

(1) X-ray diffraction / bond lengths

All six carbon–carbon bonds in benzene have the **same length, 139 pm**, which is **between** a C–C single bond (154 pm) and a C=C double bond (134 pm). The Kekulé structure would have **alternating long and short** bonds. Equal, intermediate bond lengths therefore show the electrons are **delocalised** over all six carbons. Benzene is also shown to be **planar and regular hexagonal**.

(2) Enthalpy change of hydrogenation

Cyclohexene + H ₂ → cyclohexane	$\Delta H = -120 \text{ kJ mol}^{-1}$
Expected for "cyclohexa-1,3,5-triene" = 3 × (-120)	= -360 kJ mol ⁻¹
Actual for benzene	$\Delta H = -208 \text{ kJ mol}^{-1}$
Difference = 152 kJ mol ⁻¹	

Benzene releases **less energy** than expected, so it is **152 kJ mol⁻¹ more stable** than the theoretical Kekulé structure. This extra stability is the **delocalisation (resonance) energy**, caused by the delocalisation of the π electrons.

(Different papers quote slightly different values — 150, 152 or 208 vs 360. Always use the numbers given in the question.)

Interpreting a table of dienes (Jan 2026 Q19): cyclohexa-1,4-diene gives $-240 \approx 2 \times (-120)$, because the two double bonds are **isolated** and cannot interact. Cyclohexa-1,3-

diene gives -232 , slightly less exothermic than -240 , because the two double bonds are **conjugated**, giving a **small amount of delocalisation** and extra stability. Benzene at -208 is much less exothermic still, showing **full delocalisation** over six carbons.

(3) Chemical reactivity

Benzene **does not decolourise bromine water** at room temperature, whereas an alkene does. Benzene undergoes **substitution**, not addition, and needs a **catalyst and/or heat**. The delocalised ring has a **lower electron density** than an isolated $C=C$, so it **cannot polarise a bromine molecule**; the delocalisation also makes benzene thermodynamically more stable, so addition (which would destroy the delocalisation) is unfavourable.

(4) Isomer count

Only **one** 1,2-disubstituted isomer of benzene exists. The Kekulé structure predicts **two** (one with a double bond between the substituted carbons, one with a single bond), so the fact that only one is found supports delocalisation.

10.3 Bond enthalpy calculations

Jan 2025 Q16a uses a Hess cycle to obtain a **mean C–C bond enthalpy in benzene** and asks why the value is not exact. Answer: the bonds in benzene are **not simple single or double bonds** — they are **intermediate**, and mean bond enthalpies are **averages taken over many different compounds**, so they do not apply exactly to the delocalised system.

11. Electrophilic substitution — the mechanisms

Every paper contains at least one mechanism worth 3–5 marks.

11.1 The general mechanism (and where the marks are)

- Step 1 The electrophile E^+ is generated
- Step 2 A curly arrow from **INSIDE** the ring (from the circle) to E^+
→ forms the unstable intermediate:
- a horseshoe covering **AT LEAST THREE** carbons (never all six)
 - a $+$ charge inside or just above the horseshoe
 - the sp^3 carbon carries **BOTH** H and E
- Step 3 A curly arrow from the C–H bond **BACK INTO** the ring
→ restores the delocalised system, releasing H^+
- Step 4 The catalyst is regenerated

⚠ The four errors mark schemes list explicitly:

1. Curly arrow starting **on or outside** the hexagon instead of from inside the circle — **rejected**.
2. Curly arrow going **to a lone pair** on the electrophile rather than to the atom — **rejected**.
3. **Missing the + or δ^+ on the electrophile** — **rejected**.
4. Horseshoe covering **all six** carbons, or fewer than three, or not facing the sp^3 carbon — **rejected**.

Also: the curly arrow in step 3 must start from the **C–H bond**, not from the H atom.

11.2 The five reactions

(a) Nitration

Reagents: concentrated HNO_3 + concentrated H_2SO_4
 Conditions: $50\text{--}60^\circ\text{C}$ (below 55°C to avoid further substitution)
 Electrophile: NO_2^+ (the nitronium ion)

$\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + 2\text{HSO}_4^- + \text{H}_3\text{O}^+$
 (or, simplified: $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$)
 Regeneration: $\text{H}^+ + \text{HSO}_4^- \rightarrow \text{H}_2\text{SO}_4$
 Product: nitrobenzene, $\text{C}_6\text{H}_5\text{NO}_2$

(b) Halogenation

Reagents: Br_2 (or Cl_2) with a halogen carrier: FeBr_3 , AlBr_3 , AlCl_3 , or Fe
 Conditions: room temperature or warm, anhydrous
 Electrophile: Br^+ (or a polarised $\text{Br}^{\delta+}\text{--Br}^{\delta-}\cdots\text{FeBr}_3$)

$\text{Br}_2 + \text{FeBr}_3 \rightarrow \text{Br}^+ + \text{FeBr}_4^-$
 Regeneration: $\text{FeBr}_4^- + \text{H}^+ \rightarrow \text{FeBr}_3 + \text{HBr}$

(c) Friedel–Crafts alkylation

Reagents: chloroalkane RCl + AlCl_3 (anhydrous)
 Conditions: reflux, anhydrous
 Electrophile: R^+ (a carbocation)

$\text{CH}_3\text{Cl} + \text{AlCl}_3 \rightarrow \text{CH}_3^+ + \text{AlCl}_4^-$
 Regeneration: $\text{AlCl}_4^- + \text{H}^+ \rightarrow \text{AlCl}_3 + \text{HCl}$

(d) Friedel–Crafts acylation — the route to aromatic ketones

Reagents: acyl chloride $\text{RCOCl} + \text{AlCl}_3$ (anhydrous)

Conditions: reflux, anhydrous

Electrophile: RCO^+ (an acylium ion)

$\text{CH}_3\text{COCl} + \text{AlCl}_3 \rightarrow \text{CH}_3\text{CO}^+ + \text{AlCl}_4^-$

Product with benzene: phenylethanone $\text{C}_6\text{H}_5\text{COCH}_3$

⚠ MCQ trap: to make phenylethanone from benzene you need **ethanoyl chloride with AlCl_3** — not ethanal, and **not** with a platinum catalyst (June 2019 Q10, June 2020 Q3).

(e) Sulfonation

Reagents: fuming sulfuric acid (H_2SO_4 containing dissolved SO_3),
or concentrated H_2SO_4 under reflux for several hours

Electrophile: SO_3

Product: benzenesulfonic acid, $\text{C}_6\text{H}_5\text{SO}_3\text{H}$

11.3 Free-radical substitution of the side chain

Methylbenzene + Cl_2 in UV light substitutes in the **side chain**, not the ring, because the mechanism is **free-radical** (initiated by homolytic fission of Cl_2 by UV), giving $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ and then $\text{C}_6\text{H}_5\text{CHCl}_2$, $\text{C}_6\text{H}_5\text{CCl}_3$. With a **halogen carrier and no UV**, substitution occurs **in the ring** instead. (June 2020 Q4.)

11.4 Oxidation of the side chain

$\text{C}_6\text{H}_5\text{CH}_3 + \text{alkaline KMnO}_4, \text{ reflux} \rightarrow \text{C}_6\text{H}_5\text{COO}^- \rightarrow (\text{add } \text{H}^+) \rightarrow \text{C}_6\text{H}_5\text{COOH}$

Any alkyl side chain is oxidised to a **$-\text{COOH}$** group attached directly to the ring. The benzoate ion is converted to benzoic acid by **adding a dilute strong acid** (e.g. HCl or H_2SO_4).

12. Phenol

12.1 Why phenol is more reactive than benzene

The 2–3 mark answer, learned exactly:

A **lone pair of electrons on the oxygen atom** is **delocalised into the benzene ring** (it overlaps with the π system). This **increases the electron density** of the ring. So the ring is **more easily attacked by (more attractive to) electrophiles**, and it can **polarise** a

bromine molecule without a catalyst.

⚠ **Mark-scheme rejection:** "the lone pair on the O–H is delocalised" — you must say the lone pair on the **oxygen atom**.

12.2 The key reactions

Reagent	Benzene	Phenol
Bromine water	no reaction	immediate white precipitate of 2,4,6-tribromophenol ; bromine water is decolourised ; room temperature, no catalyst
Bromine	needs FeBr ₃ , heat; gives monobromobenzene	gives the trisubstituted product
Nitric acid	needs concentrated HNO ₃ + concentrated H ₂ SO ₄ , 50–60 °C	dilute HNO ₃ at room temperature ; gives a mixture of 2-nitrophenol and 4-nitrophenol
NaOH(aq)	no reaction	dissolves — phenol is weakly acidic : $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{O}^-\text{Na}^+ + \text{H}_2\text{O}$
Na₂CO₃	no reaction	no reaction (phenol is a weaker acid than carbonic acid)
Neutral FeCl₃	no reaction	purple/violet colour
Ethanoyl chloride	Friedel–Crafts (needs AlCl ₃)	esterification at the OH → phenyl ethanoate

Tests to distinguish phenol from benzene: bromine water (white precipitate / decolourised) or neutral iron(III) chloride (purple colour) or sodium hydroxide (phenol dissolves).

12.3 The * comparison question

"Compare and contrast the bromination of benzene and of phenol" appears in **Jan 2025 Q22** and **June 2026 Q15** as a 6-mark * question.

Model answer:

Both are **electrophilic substitution** reactions. **Benzene** requires a **halogen carrier**

catalyst (FeBr₃ or AlBr₃) and **heat**, and gives **monosubstitution** to form bromobenzene: $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$. **Phenol** reacts with **bromine water at room temperature with no catalyst**, and gives **trisubstitution** at the 2, 4 and 6 positions: $\text{C}_6\text{H}_5\text{OH} + 3\text{Br}_2 \rightarrow \text{C}_6\text{H}_2\text{Br}_3\text{OH} + 3\text{HBr}$. Phenol forms a **white precipitate** and **decolourises** the bromine water; benzene does not. Phenol is **more reactive because a lone pair on the oxygen is delocalised into the ring**, increasing the ring's electron density so it can **polarise the Br–Br bond** itself. In benzene the electron density is lower, so a catalyst is needed to **generate the Br⁺ electrophile** or to polarise Br₂.

PART D — ORGANIC NITROGEN CHEMISTRY

13. Amines

13.1 Classification and naming

Primary	R-NH ₂	(one carbon on N)	e.g. butylamine, phenylamine
Secondary	R ₂ NH	(two carbons on N)	e.g. diethylamine
Tertiary	R ₃ N	(three carbons on N)	e.g. trimethylamine
Quaternary	R ₄ N ⁺	(four carbons – an ion)	

13.2 Basicity — the most-tested idea in this topic

Why amines are basic: the **nitrogen atom has a lone pair of electrons** that can **accept a proton (H⁺)** / form a **dative covalent bond** with H⁺.

The order of base strength:

secondary alkylamine > primary alkylamine > ammonia > phenylamine (aromatic amine)

Why an alkylamine is a stronger base than ammonia (Oct 2024 Q15d, 3 marks):

Alkyl groups are **electron-releasing (positive inductive effect)**. So they **push electron density towards the nitrogen atom**, increasing the electron density on N. The **lone pair is therefore more available** to accept a proton, so the amine is a **stronger base**.

Why phenylamine is a weaker base than ammonia (June 2020 Q22b(i), 3 marks — the mark-scheme wording):

The **lone pair on the nitrogen is delocalised into the benzene ring** (it overlaps with the

delocalised π system). This **decreases the electron density on the nitrogen atom**. So the **lone pair is less available to accept a proton / H^+** , making phenylamine a weaker base.

Applied to a molecule with two nitrogens (Jan 2025 Q23c(i), June 2023 Q25b(ii) — procaine):

The **aliphatic (tertiary alkyl) amine nitrogen** is the **stronger base**, because the alkyl groups **push electron density onto it**, so its lone pair is readily available. The **aromatic amine nitrogen** attached to the benzene ring is **weaker**, because its lone pair is **delocalised into the ring**. So the **aliphatic nitrogen is protonated** when procaine reacts with 1 mol of HCl.

Why an amide is not basic: the nitrogen lone pair is **delocalised onto the adjacent $C=O$ group**, so it is not available to accept a proton. (This is why caffeine is a very weak base — June 2021 Q24a(ii).)

13.3 Preparation of amines

Route	Reagents and conditions	Notes
Halogenoalkane + ammonia	excess concentrated ethanolic NH_3 , heat in a sealed tube (under pressure)	Nucleophilic substitution. Excess NH_3 minimises further substitution. Gives a mixture of 1°, 2°, 3° amines and the quaternary salt
Nitrile reduction	$LiAlH_4$ in dry ether, then dilute acid; or H_2 with a Ni/Pt catalyst, high T and p	$RCN + 4[H] \rightarrow RCH_2NH_2$ — adds one carbon
Nitroarene reduction	Sn and concentrated HCl , heat under reflux, then NaOH(aq)	$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$. The NaOH liberates the free amine from its salt
Amide reduction	$LiAlH_4$ in dry ether	$RCONH_2 + 4[H] \rightarrow RCH_2NH_2 + H_2O$

⚠ **Which amines cannot be made from a nitrile?** Reduction of a nitrile always gives a **primary amine with the $-NH_2$ on a terminal CH_2** . So an amine with the nitrogen on a **branched or secondary carbon**, or a **secondary/tertiary** amine, cannot be made this way (Oct 2021 Q14).

13.4 Reactions of amines

Reagent	Product
HCl(aq)	ammonium salt: $\text{CH}_3\text{NH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{NH}_3^+\text{Cl}^-$
HCl gas	white smoke of the solid salt $\text{CH}_3\text{NH}_3\text{Cl}$
Halogenoalkane	further substitution: $\text{C}_2\text{H}_5\text{NH}_2 + \text{C}_2\text{H}_5\text{Cl} \rightarrow (\text{C}_2\text{H}_5)_2\text{NH}_2^+\text{Cl}^-$
Acyl chloride	N-substituted amide + HCl (steamy/misty fumes): $\text{CH}_3\text{COCl} + \text{C}_6\text{H}_5\text{NH}_2 \rightarrow \text{CH}_3\text{CONHC}_6\text{H}_5 + \text{HCl}$
$\text{Cu}^{2+}(\text{aq})$	acts first as a base giving a pale blue precipitate , then as a ligand in excess giving a deep blue solution
Water	weakly alkaline: $\text{CH}_3\text{NH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{NH}_3^+ + \text{OH}^-$

Why amines dissolve in water: they form **hydrogen bonds** with water — between the **lone pair on N** and a **$\delta+$ hydrogen of water**, or between the **$\delta+$ H of N–H** and the **lone pair on O of water**. Draw the hydrogen bond as a **dotted line from the lone pair to the H**, and show the lone pair.

Naming the salt formed with an acid: $\text{CH}_3\text{NH}_3^+\text{Cl}^-$ is methylammonium chloride.

14. Amides and acyl chlorides

Acyl chloride RCOCl

Amide RCONH_2

N-substituted amide RCONHR'

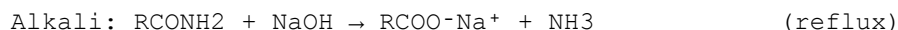
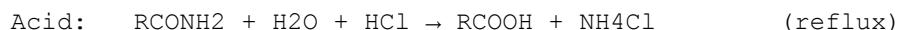
Reaction of an acyl chloride	Product
+ water	carboxylic acid + HCl
+ alcohol	ester + HCl
+ ammonia	primary amide + HCl
+ primary amine	N-substituted amide + HCl
+ phenol	phenyl ester + HCl

All are **nucleophilic addition–elimination** reactions and produce **steamy/misty white**

fumes of HCl.

Naming N-substituted amides: $\text{CH}_3\text{CONHC}_6\text{H}_5$ is **N-phenylethanamide**; $\text{C}_6\text{H}_5\text{CONHCH}_3$ is **N-methylbenzamide**. Get the order right — the acyl part gives the stem, the N-substituent is the prefix.

Hydrolysis of amides:

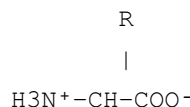


15. Amino acids, peptides and proteins

15.1 Structure and the zwitterion

General formula $\text{H}_2\text{N}-\text{CHR}-\text{COOH}$ (2-amino acids).

In the solid and at the isoelectric point the amino acid exists as a **zwitterion**:



The **–COOH group donates a proton to the –NH₂ group** within the same molecule (an internal acid–base reaction).

pH	Predominant form
Low pH (acidic)	$\text{H}_3\text{N}^+-\text{CHR}-\text{COOH}$ — cation (the COO^- is protonated)
Isoelectric point	$\text{H}_3\text{N}^+-\text{CHR}-\text{COO}^-$ — zwitterion , net charge zero
High pH (alkaline)	$\text{H}_2\text{N}-\text{CHR}-\text{COO}^-$ — anion (the NH_3^+ is deprotonated)

15.2 The two properties this explains

High melting temperature (a repeated question):

Amino acids exist as **zwitterions**, which are **ions**. There are strong **ionic attractions (electrostatic forces) between the oppositely charged groups** of neighbouring zwitterions. A large amount of energy is needed to break these, so the melting point is high. **The main forces broken on melting are ionic bonds** — not hydrogen bonds, not

London forces.

High solubility in water:

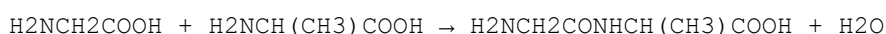
The zwitterion is **polar/ionic**, so it forms strong **ion–dipole interactions** and **hydrogen bonds** with water molecules. The energy released on hydration compensates for breaking the ionic attractions in the lattice.

15.3 Optical activity

- An amino acid is **chiral** if the α -carbon carries **four different groups**.
- **Glycine (R = H) is NOT chiral** — the α -carbon carries two hydrogens. Every other natural amino acid is.
- Chiral molecules **rotate the plane of plane-polarised light** in opposite directions; the two forms are **non-superimposable mirror images (enantiomers)**.
- **Drawing the other optical isomer:** reflect the structure in the mirror line, keeping the same four groups. Use **wedge and dashed bonds**. Do **not** simply swap two labels without redrawing.

15.4 Peptides and proteins

Peptide (amide) link: $-\text{CO}-\text{NH}-$, formed by a **condensation** reaction between the $-\text{COOH}$ of one amino acid and the $-\text{NH}_2$ of another, losing **H_2O** .



⚠ MCQ trap: the link is $-\text{CO}-\text{NH}-$ (an amide). Structures written $-\text{COO}-\text{NH}-$ are **wrong** (June 2019 Q13c).

Two amino acids give two different dipeptides, depending on which $-\text{COOH}$ reacts.

Hydrolysis of a protein/dipeptide: heat under **reflux** with **6 mol dm^{-3} hydrochloric acid** (or with NaOH , then acidify). Acid hydrolysis gives the amino acids as their **cations**.

Detection on a chromatogram: spray with **ninhydrin** (gives purple spots) — amino acids are colourless. UV light can be used if the plate is fluorescent.

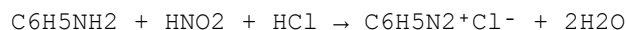
Protein structure levels: primary = the sequence of amino acids; secondary = α -helix and β -pleated sheet held by **hydrogen bonds** between $\text{C}=\text{O}$ and $\text{N}-\text{H}$; tertiary = the overall 3D fold, held by hydrogen bonds, ionic attractions, London forces and **disulfide (S–S) bridges**.

16. Diazonium salts and azo dyes

16.1 Making the diazonium salt

Reagents: sodium nitrite (NaNO_2) and dilute hydrochloric acid
– these generate nitrous acid, HNO_2 , in situ

Conditions: $0-10\text{ }^\circ\text{C}$ (an ice bath); often quoted as "below $10\text{ }^\circ\text{C}$ " or "about $5\text{ }^\circ\text{C}$ "



Why $0-10\text{ }^\circ\text{C}$? (2 marks — both halves needed)

Above $10\text{ }^\circ\text{C}$ the **benzenediazonium ion decomposes**, giving phenol and nitrogen gas.
Below about $5\text{ }^\circ\text{C}$ the reaction is **too slow**.

⚠ **MCQ trap:** the reason is **not** that the reaction is highly exothermic or that the ring would be nitrated (Oct 2019 Q9).

16.2 Coupling to make an azo dye

The diazonium ion is a weak electrophile and couples with an **activated** aromatic ring:

- with **phenol in alkaline solution** (NaOH), giving an orange/yellow azo dye;
- with an **aromatic amine**, giving a yellow/red azo dye.

Coupling occurs at the position **para to the OH or NH_2 group**.

16.3 Why azo compounds are coloured

The molecule contains an **extended delocalised electron system** (the two benzene rings joined through the **$-\text{N}=\text{N}-$** group). The energy gap between the delocalised levels corresponds to **visible light**, so part of the visible spectrum is **absorbed**; the **complementary colour** is transmitted and seen. The larger the delocalised system, the **smaller the energy gap** and the longer the wavelength absorbed.

Geometric isomerism: azo compounds show *E-Z* isomerism about the **$\text{N}=\text{N}$ double bond**, because there is **restricted rotation** about it and each nitrogen carries **two different groups** (a substituent and a lone pair).

Why azo dyes make good dyes: they are stable, strongly coloured (intense), available in a wide range of colours, and can be made to bond to fibres.

17. Polymers

17.1 The two classes

	Addition	Condensation
Monomers	alkenes (C=C)	two functional groups per monomer
Small molecule lost	none	H ₂ O or HCl
Examples	poly(ethene), poly(propene), poly(phenylethene), PVC, poly(propenamide)	polyesters, polyamides, proteins

📌 **Definition of a condensation polymer:** *a polymer formed when many monomers join together with the elimination of a small molecule, usually water (or HCl), at each link.*

17.2 Polyesters

Made from a **diol + dicarboxylic acid** (loses H₂O) or a **diol + diacyl dichloride** (loses HCl).
Link = **–COO–**.

Terylene / PET: benzene-1,4-dicarboxylic acid + ethane-1,2-diol

17.3 Polyamides

Made from a **diamine + dicarboxylic acid/diacyl dichloride**, or from a single **amino acid**.
Link = **–CONH–**.

Nylon 6,6: hexanedioic acid HOOC(CH₂)₄COOH + 1,6-diaminohexane H₂N(CH₂)₆NH₂

Nylon 6: from caprolactam, by ring-opening polymerisation

Kevlar: benzene-1,4-dicarboxylic acid + 1,4-diaminobenzene

Deducing the monomers from a repeat unit: break each **–CONH–** or **–COO–** link; add **–OH** to the carbonyl carbon and **–H** to the N or O.

Why polyamides have higher melting temperatures than polyalkenes (Oct 2023 Q20c(ii)):

Polyamides have **polar N–H and C=O groups**, so **hydrogen bonds** form **between the chains**. Polyalkenes have only **London forces** between chains. Hydrogen bonds are

stronger, so **more energy is needed** to separate the chains and the melting temperature is higher.

17.4 Water-soluble and specialist polymers

- **Poly(ethenol)** dissolves in water because it has many **–OH groups** which form **hydrogen bonds** with water.
- **Polyacrylamide (PAM)** absorbs water because its **–CONH₂ groups hydrogen bond** to water molecules — draw the hydrogen bond as a dotted line from a lone pair on the water oxygen to the amide N–H (or vice versa).
- **Polyacrylic acid above pH 8:** the **–COOH groups are deprotonated to –COO[–]**, so the intramolecular hydrogen bonds are lost and the like-charged groups **repel** each other, forcing the chain into an **open coil**.
- **Which monomers cannot form a polymer?** A monomer must have **two** reactive sites. A monoamine, a monoalcohol or a mono-acid cannot polymerise (June 2019 Q11).

PART E — FURTHER ORGANIC SYNTHESIS

18. Grignard reagents

Formation: $R-Br + Mg \rightarrow RMgBr$ in DRY ether, under reflux, anhydrous conditions

The Grignard carbon is nucleophilic (it carries a δ^- charge).

Reagent	Product after hydrolysis with dilute acid
CO ₂ (dry ice)	carboxylic acid RCOOH — adds one carbon
Methanal HCHO	primary alcohol RCH ₂ OH
Any other aldehyde RCHO	secondary alcohol
Ketone R ₂ CO	tertiary alcohol
Water	alkane (this is why the apparatus must be dry)

⚠ **MCQ:** to obtain a **secondary** alcohol you need an **aldehyde other than methanal** — i.e.

ethanal (June 2020 Q11).

Worked route (Jan 2025 Q19 — ethanoic acid to propanoic acid):

Step 1 $\text{CH}_3\text{COOH} + \text{LiAlH}_4$ in dry ether, then dilute acid $\rightarrow \text{CH}_3\text{CH}_2\text{OH}$
Step 2 PCl_5 (or PBr_3 , or HBr , or conc. HCl/ZnCl_2) $\rightarrow \text{CH}_3\text{CH}_2\text{Cl}$ (compound X)
Step 3 Mg in dry ether, reflux $\rightarrow \text{CH}_3\text{CH}_2\text{MgCl}$
Step 4 CO_2 then dilute acid $\rightarrow \text{CH}_3\text{CH}_2\text{COOH}$

19. Nitriles and chain extension

Halogenoalkane + KCN in ethanol, heat under reflux $\rightarrow$ nitrile (nucleophilic subs)
 $\text{CH}_3\text{I} + \text{KCN} \rightarrow \text{CH}_3\text{CN} + \text{KI}$ — adds one carbon

Carbonyl + HCN (KCN + dilute acid) $\rightarrow$ hydroxynitrile (nucleophilic add)
 $\text{CH}_3\text{COCH}_3 + \text{HCN} \rightarrow (\text{CH}_3)_2\text{C}(\text{OH})\text{CN}$

Nitrile reactions:

Acid hydrolysis: $\text{CH}_3\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{COOH} + \text{NH}_4\text{Cl}$ (reflux with dilut
Alkaline hydrolysis: $\text{CH}_3\text{CN} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COO}^-\text{Na}^+ + \text{NH}_3$
then add dilute acid to obtain the carboxylic acid
Reduction: $\text{CH}_3\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ (LiAlH_4 , or H_2/Ni)

Why acid must be added after alkaline hydrolysis (June 2019 Q14, Jan 2026 Q23a(iii)):

Alkaline hydrolysis produces the **carboxylate ion / sodium salt**, not the acid. The acid **protonates the carboxylate ion** to form the carboxylic acid. (**Not** "to neutralise the sodium hydroxide" and **not** "to catalyse the reaction".)

Why a hydroxynitrile from a ketone/aldehyde is not a single isomer (June 2024 Q17b, 3 marks):

The carbonyl carbon is **planar** (trigonal planar, sp^2). The **CN^- nucleophile can attack from either side** of the plane with **equal probability**. So **two enantiomers are formed in equal amounts** — a **racemic mixture**, which is optically inactive.

20. The reagent summary table (learn this)

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Conversion	Reagents and conditions
Alkene → alkane	H ₂ , Ni catalyst, heat
Alkene → dihalogenoalkane	Br ₂ or Cl ₂ , room temperature (decolourises Br ₂ water)
Halogenoalkane → alcohol	NaOH(aq), reflux
Halogenoalkane → nitrile	KCN in ethanol, reflux
Halogenoalkane → amine	excess ethanolic NH ₃ , heat under pressure
Halogenoalkane → alkene	ethanolic KOH, reflux (elimination)
Alcohol → halogenoalkane	PCl ₅ / PCl ₃ / SOCl ₂ / conc. HCl+ZnCl ₂ / HBr (from NaBr + H ₂ SO ₄)
1° alcohol → aldehyde	K ₂ Cr ₂ O ₇ / dilute H ₂ SO ₄ , distil off immediately
1° alcohol → carboxylic acid	K ₂ Cr ₂ O ₇ / dilute H ₂ SO ₄ , heat under reflux
2° alcohol → ketone	K ₂ Cr ₂ O ₇ / dilute H ₂ SO ₄ , reflux
Alcohol → alkene	conc. H ₂ SO ₄ or H ₃ PO ₄ , heat (dehydration)
Carbonyl → alcohol	NaBH ₄ in aqueous solution, or LiAlH ₄ in dry ether
Carbonyl → hydroxynitrile	HCN / KCN with dilute acid
Carboxylic acid → 1° alcohol	LiAlH ₄ in dry ether, then dilute acid
Carboxylic acid → acyl chloride	PCl ₅ / SOCl ₂
Carboxylic acid → ester	alcohol + conc. H ₂ SO ₄ catalyst, reflux
Nitrile → 1° amine	LiAlH ₄ in dry ether, or H ₂ /Ni
Nitrile → carboxylic acid	dilute HCl, reflux (or NaOH then acidify)
Nitrobenzene → phenylamine	Sn + conc. HCl, reflux, then NaOH(aq)
Phenylamine → diazonium salt	NaNO ₂ + dilute HCl, 0–10 °C
Amine → N-substituted	

amide	acyl chloride, room temperature
Benzene → nitrobenzene	conc. HNO ₃ + conc. H ₂ SO ₄ , 50–60 °C
Benzene → halogenoarene	Br ₂ with FeBr ₃ (or Cl ₂ with AlCl ₃), warm
Benzene → alkylbenzene	RCI + AlCl ₃ , reflux
Benzene → aromatic ketone	RCOCl + AlCl ₃ , reflux
Alkylbenzene → benzoic acid	alkaline KMnO ₄ , reflux, then acidify

Tests you must know:

Test	Positive result	Indicates
Bromine water	decolourised	C=C
Bromine water	white precipitate	phenol
Neutral FeCl ₃	purple	phenol
2,4-DNPH (Brady's reagent)	orange/yellow precipitate	aldehyde or ketone (C=O)
Tollens' reagent, warm	silver mirror	aldehyde
Fehling's/Benedict's, warm	brick-red precipitate	aldehyde
Iodine + NaOH, warm	pale yellow precipitate (CHI ₃ , antiseptic smell)	CH ₃ CO– group or CH ₃ CH(OH)–
Na ₂ CO ₃ (aq)	effervescence, CO ₂	carboxylic acid
Sodium metal	effervescence, H ₂	–OH (alcohol, acid, phenol)
PCl ₅	steamy/misty fumes of HCl	–OH (alcohol, acid)
Damp red litmus	turns blue	amine
Warm with NaOH then acidify, add AgNO ₃	precipitate (white/cream/yellow)	halogenoalkane
Ninhydrin	purple spots	amino acid

PART F — ANALYSIS AND STRUCTURE DETERMINATION

21. Mass spectrometry

- The **molecular ion peak (M^+)** is the peak at the **highest m/z** (ignoring isotope peaks) and gives the **relative molecular mass**.
- **$M+1$ peak:** caused by the **1.1 % of carbon that is ^{13}C** . Its relative height $\approx 1.1 \times$ (number of carbon atoms) % of the M^+ peak.
- **Chlorine:** $^{35}\text{Cl} : ^{37}\text{Cl} \approx 3 : 1$, giving M and $M+2$ in the ratio **3 : 1**.
- **Two chlorines:** $M : M+2 : M+4 = 9 : 6 : 1$.
- **Bromine:** $^{79}\text{Br} : ^{81}\text{Br} \approx 1 : 1$, giving M and $M+2$ in the ratio **1 : 1**.

Worked (June 2024 Q15d — 2,6-dichlorophenol, peaks at 162, 164, 166):

The compound contains **two chlorine atoms**. Chlorine has two isotopes, ^{35}Cl and ^{37}Cl , in a 3:1 ratio, so the combinations $^{35}/^{35}$, $^{35}/^{37}$ and $^{37}/^{37}$ occur in the ratio **9 : 6 : 1**. The corresponding peaks at 163, 165 and 167 are the **$M+1$ peaks**, due to the presence of ^{13}C .

Common fragment ions:

m/z 15 CH_3^+	m/z 29 C_2H_5^+ or CHO^+	m/z 43 CH_3CO^+ or C_3H_7^+
m/z 57 $\text{C}_2\text{H}_5\text{CO}^+$ or C_4H_9^+		m/z 77 C_6H_5^+
m/z 105 $\text{C}_6\text{H}_5\text{CO}^+$	m/z 91 $\text{C}_6\text{H}_5\text{CH}_2^+$ (tropylium)	

Fragment ions arise from **homolytic or heterolytic fission** of the molecular ion; only the **positively charged** fragment is detected.

22. Infrared spectroscopy

Use the Data Booklet ranges — but know the key ones:

Bond	Wavenumber / cm^{-1}	Appearance
O–H (carboxylic acid)	2500–3300	very broad

O–H (alcohol, phenol)	3200–3600	broad
N–H (amine, amide)	3300–3500	often a double peak for a primary amine
C–H	2850–3100	
C≡N (nitrile)	2220–2260	sharp, in an otherwise empty region
C=O	1680–1750	strong, sharp
C=C	1620–1680	weak
C–O	1000–1300	

Exam technique: quote **both the bond and the wavenumber range from the Data Booklet** — mark schemes require the range, not just the identification. The **absence** of a peak is as much evidence as its presence: no broad 2500–3300 peak means **no carboxylic acid**.

23. NMR spectroscopy

23.1 Carbon-13 NMR

- **Number of peaks = number of carbon environments.**
- Use **symmetry** to count: in methylcyclohexane there are **five** environments (the CH₃, C1, and then C2/C6, C3/C5, C4 as three pairs).
- Chemical shifts from the Data Booklet: C=O ~190–220; aromatic C 110–160; C–O 50–90; C–C 5–55.

23.2 Proton (¹H) NMR

Feature	What it tells you
Number of peaks	number of hydrogen environments
Relative peak areas (integration)	the ratio of the numbers of hydrogens in each environment
Chemical shift δ	the type of environment (use the Data Booklet table)
	the number of hydrogens on adjacent carbon atoms : $n + 1$

Splitting pattern	peaks for n neighbours
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The n+1 rule:

0 neighbours → singlet	1 → doublet	2 → triplet	
3 → quartet	4 → quintet	5 → sextet	6 → heptet

Worked: propan-1-ol CH3CH2CH2OH gives **one singlet (OH), two triplets and one sextet** — the CH₃ has 2 neighbours (triplet), the middle CH₂ has 5 neighbours (sextet), the CH₂-O has 2 neighbours (triplet), and the OH is a singlet. Propan-2-ol (CH3)2CHOH gives **one singlet (OH), one doublet (the 6 CH₃ protons) and a heptet (the CH)**.

O–H and N–H protons:

- Give a **broad singlet** and are **not split**, and do **not split** neighbouring protons, because of **rapid proton exchange**.
- They **disappear** (or move) when **D₂O is added**, because the H is exchanged for D. Running a spectrum with and without D₂O identifies the OH/NH peak.

TMS (tetramethylsilane) as the reference standard: it is **inert, non-toxic, volatile** (easily removed), gives a **single sharp peak** from 12 equivalent hydrogens, and that peak is **upfield of almost all other signals** (defined as $\delta = 0$).

Solvents: CDCl₃ or CCl₄ — they contain **no hydrogen atoms**, so they give no peaks that would obscure the spectrum.

23.3 The structure-determination routine

When a question gives you several pieces of data, work in this order:

1. MASS SPECTRUM → M_r from the molecular ion; look for Cl/Br isotope patterns
 2. COMBUSTION DATA / % composition → empirical formula → molecular formula
 3. DEGREES OF UNSATURATION: for C_xH_yO_z, $\text{DoU} = (2x + 2 - y) / 2$
(a benzene ring = 4; C=O = 1; C=C = 1; ring = 1)
 4. INFRARED → which functional groups are PRESENT and ABSENT
 5. CHEMICAL TESTS → confirm the functional groups
 6. ¹³C NMR → number of carbon environments (symmetry)
 7. ¹H NMR → number of H environments, ratio, and neighbours
 8. Assemble, then CHECK your structure against EVERY piece of data
-

24. Chromatography and separation

Technique	Principle	Notes
TLC / paper	different affinities for the stationary and mobile phases	$R_f = \frac{\text{distance moved by spot}}{\text{distance moved by solvent front}}$
Gas chromatography (GC)	volatile components separated by boiling point and affinity for the stationary phase	gives retention times ; compare with standards
HPLC	for non-volatile or heat-sensitive compounds	high pressure forces the liquid mobile phase through
GC-MS / LC-MS	the technique used with GC or HPLC to identify each separated component	gives the mass spectrum of each peak, so components can be identified from their M_r and fragmentation

Why a mass spectrometer is coupled to GC or HPLC: chromatography alone gives only retention times, which can be ambiguous; the **mass spectrum identifies** each component definitively.

Solvent extraction: shaking a mixture with a solvent in which the desired compound is **much more soluble** than the impurities (and which is **immiscible** with the original solvent), then separating the two layers in a **separating funnel**.

25. Practical techniques

Technique	Purpose and key points
Heating under reflux	heating a reaction for a long time without losing volatile substances — vapour condenses and returns. The condenser must be open at the top and water enters at the bottom .
Distillation	separating a product from the mixture as it forms (e.g. an aldehyde from a primary alcohol), or purifying by boiling point
Recrystallisation	purifying a solid : dissolve in the minimum volume of hot solvent , filter hot to remove insoluble impurities, cool to crystallise, filter under reduced pressure (Büchner), wash with a little cold solvent , dry

Melting temperature	purity check: a pure substance melts sharply at the correct temperature ; impurities lower and broaden the melting range
Boiling under reduced pressure	for compounds that decompose at their normal boiling point (e.g. salicylaldehyde at 197 °C)
Separating funnel	separating immiscible liquid layers; run off the lower layer
Drying agent	anhydrous MgSO ₄ or Na ₂ SO ₄ added to the organic layer, then filtered off

Why the melting point of an impure sample is lower: impurities **disrupt the crystal lattice**, so less energy is needed to break it, and melting occurs **over a range** rather than sharply.

Percentage yield and atom economy:

$$\% \text{ yield} = (\text{actual moles of product} \div \text{theoretical moles}) \times 100$$

$$\text{atom economy} = (\text{M}_r \text{ of desired product} \div \text{total M}_r \text{ of all products}) \times 100$$

Limiting reagent: divide the moles of each reactant by its coefficient; the **smallest** value is the limiting reagent, and the theoretical yield is calculated from it.

Uncertainty:

$$\% \text{ uncertainty} = (\text{uncertainty} \div \text{measurement}) \times 100$$

A burette reading requires TWO readings, so use $2 \times 0.05 = \pm 0.10 \text{ cm}^3$

Total % uncertainty = sum of the individual % uncertainties

The number of **significant figures** in the final answer should reflect the total percentage uncertainty — quoting 95.863 % when the total uncertainty is about 1 % is **not appropriate**; three significant figures is (June 2021 Q18c).

Factors chemists consider when choosing a synthesis route (June 2024 Q15c): **yield, atom economy, cost and availability of reagents, energy requirements, number of steps, hazards/safety, waste and environmental impact, ease of separating the product.**

PART G — EXAM TECHNIQUE

26. The asterisked (*) question

There are one to three per paper. Marks are awarded for **indicative content plus structure and linked reasoning**.

The two consequences:

- A list of disconnected facts will not reach full marks even if every fact is correct.
- Connectives are worth marks: *because, so, this means that, therefore, whereas, in contrast, however, as a result*.

For a "compare and contrast" * question, structure it as:

1. Start with the SIMILARITIES (both are electrophilic substitution; both...)
2. Then the DIFFERENCES, point by point, comparing directly
(benzene requires ____ WHEREAS phenol requires ____)
3. Give EQUATIONS where the question asks for them
4. Finish with the EXPLANATION of why the difference exists

The * topics that have appeared:

Topic	Papers
Compare and contrast heterogeneous and homogeneous catalysis	Jan 2021
Compare and contrast bromination of benzene and phenol	Jan 2025, June 2026
Compare and contrast benzene and cyclohexene bromination	June 2021
Explain why complexes are coloured / different colours	Jan 2026, Oct 2024, Jan 2024, Jan 2022, Aug 2026
Evidence for delocalisation in benzene	June 2023, Oct 2022
Explain how a fuel cell works, with advantages and disadvantages	June 2024
Reactions of nickel(II) sulfate compared with another salt	June 2022
Assess a teacher's/student's statement about a	

titration	Oct 2021
Reduction of vanadium(V) step by step	Jan 2023
Deduce a structure from NMR and other data	June 2021
Why Cr^{6+} aqua ions do not exist	June 2019

Note the pattern: *colour of complexes* and *comparisons of aromatic reactivity* together account for more than half of all * questions. Have both fully rehearsed.

27. Mark-scheme phrase bank

Learn these verbatim; the mark schemes reject close paraphrases.

Transition metals

- "A transition element forms **at least one stable ion with a partially filled d subshell.**"
- "The **ligands cause the d orbitals to split** into two energy levels."
- "**Light is absorbed**, promoting an **electron from the lower to the higher** energy level."
- "The light **not absorbed is transmitted/reflected** — the **complementary colour** is seen."
- "The ion has a **full 3d subshell ($3d^{10}$)**, so **no electron can be promoted.**"
- "Different ligands cause **different splitting of the d orbitals**, so a **different frequency of light is absorbed.**"
- "A **ligand** is a species that **donates a lone pair of electrons** to form a **dative covalent bond.**"
- "There is an **increase in the number of particles**, so **ΔS_{system} is positive.**"

Electrochemistry

- " E°_{cell} is **directly proportional to ΔS_{total} and to $\ln K$.**"
- "A **positive E°_{cell}** means the reaction is **thermodynamically feasible.**"
- "The reaction may still not occur because the **activation energy is very high** / the conditions are **not standard.**"
- "The salt bridge **completes the circuit** and allows **ions to flow**, and does **not react**

with the solutions."

Arenes and phenol

- "The **p orbitals overlap sideways** to form a **delocalised π system** above and below the ring."
- "All the C–C bonds are the **same length, intermediate** between C–C and C=C."
- "Benzene is **more stable than the Kekulé structure by ____ kJ mol⁻¹**."
- "A **lone pair on the oxygen atom is delocalised into the ring**, increasing the **electron density** of the ring."
- "Benzene **cannot polarise** the bromine molecule, so a **halogen carrier** is needed to generate the electrophile."

Amines and amino acids

- "Alkyl groups are **electron-releasing**, so the **lone pair on nitrogen is more available**."
- "The **lone pair on nitrogen is delocalised into the ring**, so it is **less available to accept a proton**."
- "Amino acids exist as **zwitterions**, so there are strong **ionic attractions** between molecules."

Catalysis

- "Both catalysts provide an **alternative route with a lower activation energy**."
- "A **heterogeneous** catalyst is in a **different phase** from the reactants."
- "Reactants are **adsorbed** onto **active sites**, the **bonds are weakened**, reaction occurs, and the products are **desorbed**."
- "The transition metal ion is **oxidised then reduced back** to its original oxidation state."

28. Section C technique

Section C is a single **20-mark, context-based** question built on a passage about a real substance (coffee, iron, dyes, local anaesthetics, liquitabs, cochineal, ketones).

How it is constructed:

- 60–70 % of the marks are **standard Unit 5 content** in an unfamiliar dress.

- 20–30 % are **synoptic** — moles, rate equations and half-lives (Unit 4), bonding and shapes (Unit 1/2), functional group chemistry (Unit 2/4).
- A few marks are for **reading the passage** and extracting information.

Technique:

1. **Read the passage once, quickly**, then go to the questions. Do not annotate heavily first — the questions tell you what matters.
2. When a structure is given, **count the functional groups and label them on the diagram** before you start. Most sub-questions turn on identifying one of them.
3. **Molecular formula from a skeletal structure:** count carbons at every line end and junction, then add hydrogens to bring each carbon to four bonds. Count the heteroatoms separately. Check twice.
4. For **half-life questions**, use $\text{after } n \text{ half-lives, amount} = \text{initial} \times (\frac{1}{2})^n$, then solve for n and multiply by $t_{1/2}$.
5. For **"suggest"** questions there is usually more than one acceptable answer — write a sensible chemical idea rather than leaving it blank.

A worked half-life example (June 2023 Q25f): articaïne, $t_{1/2} = 20 \text{ min}$, 100 mg, after 4 hours = 240 min = **12 half-lives**. $100 \times (0.5)^{12} = 0.0244 \text{ mg} = 24.4 \text{ }\mu\text{g}$.

29. Per-paper question map

Paper	Section B and C contents
2019 Jun (WCH05)	Chromium: E°cell for corrosion, reduction to Cr(II), colours, complexes with HCl and NH ₃ , chromate/dichromate, why Cr ⁶⁺ aqua ion cannot exist*, dichromate–thiosulfate titration · organic nitrogen · Section C
2019 Oct (WCH05)	Manganese: reduction by a metal, cell set-up, K ₂ MnO ₄ preparation, disproportionation feasibility · copper and zinc: configurations, ionisation energies*, why Zn ²⁺ colourless*, ammonia reactions, amphoteric zinc, brass analysis by iodine–thiosulfate* · delocalisation: carboxylate, benzene, phenol C–O bond*, nitration · amino acids: solubility, ninhydrin, dipeptide deduction · Section C: ketones — Friedel–Crafts mechanism, aldol, hydroxy acids, combustion analysis, raspberry ketone NMR
	Electrochemical cells: SHE apparatus, cell diagram half-equations, Nernst calculation · aromatic amines and amino acids: nitration mechanism errors,

2020 Jun	phenylamine basicity, azo dye synthesis, optical isomers · gas volume combustion, three-step synthesis · transition metals: Fe(III) reactions, ligand types
2021 Jan	Copper reaction sequence (7 species), T and V colours, disproportionation of CuI, mitscherlichite water of crystallisation · butanone: mass spectrum fragments, synthesis of propan-1-ol and 2-methylbut-2-ene · dichromate/ethanol cell · gas-volume combustion · * compare Pt and V2O5 catalysis · Section C: coffee — caffeine bonding and basicity, concentration, half-life, electrophilic substitution mechanism, NMR of acetoin
2021 Jun	Manganese: disproportionation of MnO_4^{2-} , feasibility of preparing MnO_4^{2-} , steel wire titration, uncertainty and significant figures · unknown amine X: ideal gas M_r , reaction with ethanoyl chloride, empirical formula, NMR deduction* · benzene thermochemistry, bromine comparison, phenol reactivity · Section C: iron — Fe(III) stability, third ionisation energy, haemocyanin colour, colorimetry ppm, thioglycolic acid ligand, homogeneous catalysis
2021 Oct	Iron complexes A–E: hydrated salt formula, cyanide complex 3D structure, empirical formula, reaction types · benzene: Kekulé/Dewar/Ladenburg structures, X-ray evidence, enthalpy level diagram, hydrogenation of dienes, Friedel–Crafts mechanism and ^{13}C NMR · TiCl_3 : ligands, stereoisomer, nitrate titration and oxidation state, indicator* · Section C: colour — curcumin radicals, nucleophilic addition, bidentate complexes, methyl red synthesis, indigotin yield, coumarin hydrolysis
2022 Jan	Aromatic amide synthesis: mechanism errors, diazonium temperature, acyl chloride structures · * nickel(II) sulfate reactions · cinnamon oil: combustion analysis and tests · chromium quantitative analysis: colours, indicator, DPC complex, colorimetry, nickel determination
2022 Jun	(as above, plus) brass and gilding metal analysis, thermodynamic vs kinetic feasibility, malleability, Nernst equation and cell apparatus
2022 Oct	Vanadium: highest oxidation state, thiosulfate reduction, why nickel is unsuitable, ferrovanadium titration, V_2O_5 catalysis · * delocalised electron systems · coniine: ideal gas M_r , NMR of the alkyl group, stereoisomerism · pigments: viridian formula, cobalt oxide with conc. HCl, shape change · Section C: plant chemicals — salicin, salicylaldehyde, Gattermann–Koch mechanism, coumarin, salen ligand stability
2023 Jan	Compound A structure determination from analysis · vanadium oxidation states, hydrated sulfate formula, MnO_4^- ratio, stepwise reduction* · organic nitrogen: phenol nitration mechanism, synthesis of N-phenylethylethanamide, phenylalanine zwitterion, aspartame chiral centres · manganese: configurations,

	electrode potentials, MnO ₂ as oxidant and catalyst, K ₂ MnO ₄ , side-chain oxidation, Zn/ZnO cell
2023 Jun	Mercury: d-block but not transition, redox with HNO ₃ , mercury(II) fulminate, calomel electrode and SHE diagram · spinach back-titration · * evidence for delocalisation in benzene · polymers: PAM, PAA pH dependence, PVP, nylon 6,6 and nylon 6 · Section C: 'Caine' anaesthetics — functional groups, procaine basicity, lidocaine synthesis and mechanism, chirality, half-life
2023 Oct	Cells: apparatus diagram, dichromate oxidising ability, cell diagram, X ₂ O ₅ oxidation state · iron: empirical formula, persulfate catalysis, monodentate ligand, octahedral shape, haemoglobin, ethanedioate titration · benzene: bromination mechanism, bromobenzene → benzamide synthesis, semi-aromatic polyamides · Section C: aspirin — naming, esterification, yield, dissociation, NMR
2024 Jan	Kekulé vs delocalised: X-ray, 1,2-dichlorobenzene, hydrogenation enthalpies, orbital description · N-phenylethanamide synthesis: nitration mechanism, reduction, overall yield, paracetamol · gingerol/zingerone · Latimer diagram: H ₃ MnO ₄ disproportionation feasibility, KMnO ₄ standardisation titration · isomerism in complexes: hydrate isomers and colour, cis-platin and DNA, Reinecke's salt, tetrahedral optical isomers
2024 Jun	Chromium cells: electrochemical series, SHE apparatus, running the cell, choosing reductants by E°cell · 2,6-dichlorophenol synthesis: sulfonation, chlorination conditions, factors in choosing a route, chlorine isotope pattern · wastewater: precipitation, dithiocarbamate ligand, feasibility, concentration · phenylglycine: Friedel–Crafts mechanism, racemic hydroxynitrile, hydrolysis · * fuel cells · Section C: liquitabs — hazard symbols, poly(ethenol), glycerol, IR and combustion structure determination
2024 Oct	Butylamine: hydrogen bonding diagram, exposure limits, amide formation, basicity · iron supplements · cells: dichromate, cell diagram, DMFC half-equations · coloured complexes: why Fe(II) is coloured, why the two differ, thiocyanate dot-and-cross and stoichiometry, colorimetry graph, chelate effect and bond angles · Section C: raspberry ketone and ionones — synthesis mechanism, stereoisomerism, ¹³ C NMR
2025 Jan	Cells: salt bridge and half-cell contents, why not a wire, dilution effect · Grignard synthesis table · chromium: reactions 1–4, choosing a metal by E°cell, redox or not, double salt formula and water of crystallisation, ppb calculation, EDTA chelation · iron(II) sulfate: configuration, air oxidation, moss-killer titration, acidity of Fe ²⁺ · * bromination of benzene vs phenol · Section C: procaine — molecular formula, synthesis with mechanism, basicity of two nitrogens, solubility of the salt, molecule count

2025 Jun	Cell with salt bridge, cell diagram, titanium electrode, changes on running · cis-platin: shape, trans isomer, ligand exchange, DNA chelation entropy · benzene: Hess cycle bond enthalpy, C–C bond formation, chemical and physical properties · trimethylamine: skeletal structure, class, solubility, alkalinity, N-oxide, GC and HPLC · sunset yellow: diazonium reagents, structure, purification, yield · * [CuCl₄]²⁻ reactions · Section C: salicylic acid — chirality, cinnamic acid, IR, nitration mechanism, solubility
2025 Oct	Structure determination of C ₈ H ₈ O ₂ isomers by IR, MS and NMR · transition elements: vanadium reduction, V ₂ O ₅ in the Contact process, platinum in catalytic converters, Mn ²⁺ autocatalysis, Fe ³⁺ /Fe ²⁺ potential · redox titrations: pyrolusite, KMnO ₄ advantages, Winkler method, dichromate/chromate equilibrium and indicators, cerium(IV) and colour · Section C: natural colourings — carminic acid, pH-dependent colour, lycopene colorimetry, solvent extraction
2026 Jan	Friedel–Crafts acylation of methoxybenzene: limiting reagent, yield, mechanism, isomer ratio · hydrogenation enthalpies of four C ₆ molecules · cobalt complexes A–D and why they are coloured* · cerium(IV)/tin(II) titration and oxidation state · persulfate–iodide: cell diagram, half-equation, feasibility, kinetic barrier, Fe catalysis justified by E° · Section C: chain extension — nitrile and Grignard routes, rhodium catalytic cycle, oxidation state, square planar and octahedral structures, rate equation and rate-determining step
2026 Jun	Vanadium and zinc: configurations, ionisation energies, oxidation states, why V ²⁺ is coloured, why Zn ²⁺ is not · histidine: molecular formula, % composition, protonation, melting behaviour, dipeptides, cobalt complex · C ₈ H ₇ ClO isomers: Friedel–Crafts mechanism, three chemical tests, 2,4-DNPH and recrystallisation · * phenol vs benzene with bromine · manganate(VII): preparation, redox proof, two disproportionations compared, oxidation of water, side-chain oxidation

Note: questions recur heavily. The colour-of-complexes explanation, a redox titration calculation, an electrophilic substitution mechanism and an E°cell feasibility argument appear in **every single paper**.

30. The twenty most commonly lost marks

1. **Writing “d orbital” (singular)** instead of “d orbitals” when explaining splitting — automatically rejected.
2. **Saying “the d orbitals cannot be split”** for a colourless d¹⁰ ion, instead of “no electrons can be promoted”.

3. **Explaining a transition element in terms of the atom** instead of the **ion**.
 4. **Forgetting H^+ in an acidified half-cell** when listing what a half-cell must contain.
 5. **Curly arrow starting on or outside the hexagon** rather than from inside the delocalised ring.
 6. **Horseshoe covering all six carbons** in the electrophilic substitution intermediate (it must span 3–5, facing the sp^3 carbon).
 7. **Omitting the + or $\delta+$ on the electrophile**.
 8. **Saying “the lone pair on the OH is delocalised”** rather than “the lone pair on the oxygen atom”.
 9. **Not quoting oxidation numbers** when asked to prove a reaction is redox.
 10. **Using HCl instead of H_2SO_4** to acidify a manganate(VII) titration, or failing to explain why.
 11. **Confusing deprotonation with ligand exchange** when NaOH or limited NH_3 gives a precipitate.
 12. **Giving $[Cu(NH_3)_6]^{2+}$** instead of the correct $[Cu(NH_3)_4(H_2O)_2]^{2+}$.
 13. **Forgetting the $\times 10$ or $\times (250/25)$ scaling factor** in a titration calculation.
 14. **Not giving the answer to an appropriate number of significant figures** when the question asks.
 15. **Writing the peptide link as $-COONH-$** rather than $-CONH-$.
 16. **Claiming amino acids melt by breaking hydrogen bonds** rather than **ionic** attractions between zwitterions.
 17. **Saying E°_{cell} is proportional to ΔS_{system} or to K** rather than **ΔS_{total} and $\ln K$** .
 18. **Failing to mention the kinetic barrier** when a feasible reaction does not occur.
 19. **Using the diameter of the whole ion instead of the metal's own charge** when assigning oxidation states in complexes.
 20. **Leaving Section C short of time** — it is 20 marks and is always reachable if Section A is kept to 20 minutes.
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31. Formula and data summary

Given in the Data Booklet

Standard electrode potentials (both left-hand and right-hand electrode system tables) · IR absorption wavenumbers · ^1H and ^{13}C NMR chemical shift tables · the Periodic Table · common relative atomic masses · gas constant and molar volume.

NOT given — you must know

Oxidation number rules

$E^\circ_{\text{cell}} = E^\circ(\text{right}) - E^\circ(\text{left})$; positive $\Rightarrow$ feasible

$E^\circ_{\text{cell}} \propto \Delta S_{\text{total}}$ and $\propto \ln K$; $\Delta G = -nFE^\circ_{\text{cell}}$

Standard conditions: 298 K, 100 kPa, 1.00 mol dm $^{-3}$

Colour of every common aqua ion and its NaOH / NH $_3$ / HCl products

The three sentences explaining colour, and the two explaining colourlessness

Shapes: 6 $\rightarrow$ octahedral 90°; 4 $\rightarrow$ tetrahedral 109.5° or square planar 90°; 2 $\rightarrow$ line

The chelate effect argument (particle count $\rightarrow \Delta S_{\text{system}} \rightarrow \Delta S_{\text{total}}$)

Mechanisms and electrophiles for nitration, halogenation, alkylation, acylation,

The generating equations for each electrophile and each catalyst regeneration

Benzene stability: bond length 139 pm; delocalisation energy from hydrogenation e

Basicity order and its two explanations (inductive effect; delocalisation into th

Diazotisation conditions (0–10 °C) and BOTH reasons for the range

Zwitterion structures at low, iso-electric and high pH

The full reagent-and-conditions table (§20) and the chemical tests table

$n = cV/1000$; $n = m/M$; $n = V/24.0$ (r.t.p.) ; $pV = nRT$

% yield ; atom economy ; % uncertainty ; ppm and ppb conversions

$n+1$ splitting rule ; M+2 isotope patterns (3:1 for Cl, 1:1 for Br, 9:6:1 for Cl $_2$)

Degrees of unsaturation = $(2C + 2 - H)/2$

32. A revision plan

Where study hours earn the most marks, in order:

1. **The colour explanation and the aqua-ion reaction tables (§7, §8).** 15–25 marks a paper and entirely memorisable.
2. **The redox titration calculation layout (§4.2).** 5–10 marks a paper, always the same six lines.
3. **The five electrophilic substitution mechanisms (§11).** 3–5 marks a paper, and the marking points are fixed.

4. **E°cell, feasibility and the kinetic limitation (§2.6).** 8–17 marks a paper.
5. **The reagent-and-conditions table (§20).** Underpins every synthesis question and every “identify the reagent” mark.
6. **The basicity explanations and the zwitterion (§13.2, §15).** 5–10 marks a paper.
7. **Structure determination from combined spectra (§23.3).** Worth practising on whole past questions rather than fragments.

Five-step routine for every Section B question:

1. Circle the command word and the mark allocation
2. Underline every piece of data and note its units
3. For calculations: equation → substitution → answer, with units, on separate line
4. For explanations: write the number of distinct chemical statements the marks demand using the mark-scheme phrases in §27
5. Check significant figures, state symbols (if asked), and charges on every compound

Exam-day tactics.

- **Section A in 20 minutes**, no more. It is 20 marks — the same as the whole of Section C.
- Open the **Data Booklet** at the electrode potentials page before you start Section B.
- In mechanisms, **draw big**. Small diagrams lose arrow-position marks.
- Every complex ion must have its **square brackets and overall charge**.
- Every equation must **balance for atoms and for charge** — check both.
- Leave five minutes to check that every “give your answer to an appropriate number of significant figures” instruction has been obeyed.